RV College of Engineering®



Master of Computer Applications

Scheme And Syllabus Of I & IV Semester
(2026 Scheme)

B.E. Programs : CI, AS, BT, CD, CH, CS, CV, CY, EC, EE, EI, ET, IM, IS, ME.
M. Tech (13) MCA, M.Sc. (Engg.)
Ph.D. Programs : All Departments are recognized as Research Centres by VTU Except AI & AS

2026

RV College of Engineering®

NAAC GRADE A+ ACCREDITED INSTITUTION

SINCE 1963

SHAPING VISIONARIES, CRAFTING INNOVATORS



An autonomous institution affiliated with VTU, Belagavi, and approved by AICTE and UGC, New Delhi

RV SATELLITE-1 (RVSAT - 1) India's First Microbiological Payload, launched by RVCE students in December 2024

NCC - Best Institution Award of Karnataka and Goa Directorate (2019, 2022, 2023 & 2025)

AUTONOMOUS CAR India's first driverless car has been successfully tested at Bengaluru's RV College of Engineering as a part of Wipro-IISc-RVCE WIRIN (Wipro-IISc Research and Innovation Network) initiative.

₹50 Cr. MoU with TATA Technologies to establish Centre for Invention, Innovation, Incubation & Training.

MoU with Bajaj Auto Foundation on to implement the **Bajaj Engineering Skills Training (BEST) Programme**.The initiative will provide industry-focused training and hands-on learning in Electronics Manufacturing Systems (EMS).

Placements

- **₹67 LPA** Highest Package
- **250+** Recruitment Partners
- **92%** Placement Rate
- **₹11 LPA** Median Salary

Rankings

- **101-150** NIRF Rankings 2025 Engineering Category
- **1501+** Times Higher Education World University Rankings-2026
- **601+** Times Higher Education Asia University Rankings-2025
- **NBA** Eligible UG and PG programs accredited multiple times
- **IIRF 2025** Engineering ranking India National - 01 State - 01 Zone - 01
- **NPTEL** Local Chapter Ranking 2025 National - 21 State – 01 AA Rating

Research and Centres

37 Skill Labs

General Skills, Domain Skills, Interdisciplinary Skills

- Bosch Rexroth
- Mercedes Benz
- Morris Garage
- Boston AI System
- Toyota Kirloskar
- Vision Astra EV Academy

Outcome Based Education (OBE) emphasizing on experiential learning, research-based learning and skill-based learning

- **20** Centres of Excellence
- **8** Centres of Competence
- **123** Patents filed
- **98** Patents published
- **51** Patents granted
- **32** Innovative & Cultural activity teams
- **₹30 Cr.** Research and projects consultancy (last 5 years)

16 Research Centres in niche areas

- Autonomous Vehicles
- Nanomaterials and Devices
- Hydrogen and Green Technology
- Extended Reality and Visnet Computing
- Healthcare
- E-mobility
- IoT and Sensors
- Smart Antenna System
- Computational Genomics
- Quantum Computing

Programs, Intake and Staff

- **13 UG** Programs;**1420 (2025)** annual sanctioned intake
- **13 PG** Programs including MCA **354 (2025)** annual sanctioned intake
- **Ph.D.** Programs in all departments
- **500+** Faculty Members and Staff

State-of-the-Art infrastructure

- **Innovation Center**
- **Student Spaces**
- **Gymnatorium**
- **Hostels**
- **Labs & Workshops**
- **Food Court**

CET / KEA Code: **E005**
COMED-K Code: **E095**
PGCET Code (MTech): **T857**
PGCET Code (MCA): **C463**

Alumni

- **36,000 +** Alumni across the globe
- **100+** RVCE alumni in civil service

RVCE Reconnect - USA edition, our first International Alumni Meet

NUMBER OF ATTENDEES

- **650** San Jose
- **120** Seattle

Mr. Suresh Katta, RVCE alumnus & Chairman Emeritus of Saama Group, USA and **Mr. Anil Kumble,** RVCE alumnus & former Indian cricketer, were the Guests of Honour at the alumni meets held in San Jose & Seattle, USA.

Collaborations with 140+ MoUs with Industry, Institutions & Research Establishments

International universities

thws Technical University of Applied Sciences Würzburg-Schweinfurt · Fachhochschule Dortmund University of Applied Sciences and Arts · Technische Hochschule Rosenheim · HTWD Hochschule für Technik und Wirtschaft Dresden, University of Applied Sciences · The University of Iowa · FIU Florida International University · Saint Peter's University · Kennesaw State University · RWTH Aachen University · The University of Arizona

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Ph.D. Programs : All Departments are recognized as Research Centres by VTU Except AI & AS

2026

Master of Computer Applications

DEPARTMENT VISION

Pioneering in ICT Enabled Quality Education and Research with a focus on Sustainable and Inclusive Applications

DEPARTMENT MISSION

1. To adapt novel methodologies for quality education through experiential learning.
2. To empower students with continuous, holistic education, emphasizing discipline, ethics and social commitment.
3. To become a vibrant knowledge center for research and software development.
4. To continuously build capacity steering towards industry- institute collaborative research and entrepreneurial competencies.
5. To utilize and develop free and open-source software tools for sustainable and inclusive growth.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO1 Practice software engineering principles and standards to develop software to meet customer requirements across verticals

PEO2 Contribute to build sustainable and inclusive applications using mathematical, simulation and meta heuristic models

PEO3 Demonstrate entrepreneurial qualities through individual competence and team work

PEO4 Achieve successful professional career with integrity and societal commitments leading to lifelong learning

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1 Solve real world computing system problems of various industries by understanding and applying the principles of mathematics, computing techniques and business concepts

PSO2 Design, test, develop and maintain desktop, web, mobile and cross platform software applications using modern tools and technologies

GLOSSARY OF ABBREVIATIONS

1.	AS	Aerospace Engineering
2.	BS	Basic Sciences
3.	BT	Biotechnology
4.	CH	Chemical Engineering
5.	CHY	Chemistry
6.	CIE	Continuous Internal Evaluation
7.	CS	Computer Science & Engineering
8.	CV	Civil Engineering
9.	EC	Electronics & Communication Engineering
10.	EE	Electrical & Electronics Engineering
11.	ET	Electronics & Telecommunication Engineering
12.	GE	Global Elective
13.	HSS	Humanities and Social Sciences
14.	IM	Industrial Engineering & Management
15.	IS	Information Science & Engineering
16.	MA	Mathematics
17.	MBT	M. Tech in Biotechnology
18.	MCE	M. Tech. in Computer Science & Engineering
19.	MCN	M. Tech. in Computer Network Engineering
20.	MCS	M. Tech. in Communication Systems
21.	MDC	M. Tech. in Digital Communication
22.	ME	Mechanical Engineering
23.	MHT	M. Tech. in Highway Technology
24.	MIT	M. Tech. in Information Technology
25.	MMD	M. Tech. in Machine Design
26.	MPD	M. Tech in Product Design & Manufacturing
27.	MPE	M. Tech. in Power Electronics
28.	MSE	M. Tech. in Software Engineering
29.	MST	M. Tech. in Structural Engineering
30.	MVE	M. Tech. in VLSI Design & Embedded Systems
31.	N	Internship
32.	PHY	Physics
33.	SDA	Skill Development Activity
34.	SEE	Semester End Examination
35.	VTU	Visvesvaraya Technological University

POST GRADUATE PROGRAMS

Sl. No	Core Department	Program	Code
1.	BT	M. Tech in Biotechnology	MBT
2.	CS	M. Tech in Computer Science & Engineering	MCE
3.	CS	M. Tech in Computer Network Engineering	MCN
4.	CV	M. Tech in Structural Engineering	MST
5.	CV	M. Tech in Highway Technology	MHT
6.	EC	M. Tech in VLSI Design & Embedded Systems	MVE
7.	EC	M. Tech in Communication Systems	MCS
8.	EE	M. Tech in Power Electronics	MPE
9.	ET	M. Tech in Digital Communication	MDC
10.	IS	M. Tech in Software Engineering	MSE
11.	IS	M. Tech in Information Technology	MIT
12.	ME	M. Tech in Product Design & Manufacturing	MPD
13.	ME	M. Tech in Machine Design	MMD
14.	MCA	Master of Computer Applications	MCA

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MASTER OF COMPUTER APPLICATIONS

I SEMESTER MCA												
			Credit Allocation									
SL No	**Course Code**	**Course Title**	**L**	**T**	**P**	**Total Credits**	**BoS**	**Category**	**CIE Duration (H)**	**Max Marks CIE**	**SEE Duration (H)**	**Max Marks SEE**
1.	MMA211TD	Discrete Mathematics and Probability Theory	3	1	0	**4**	MA	Theory	1.5	100	3	100
2.	MCA112IA	Web Application Development	3	0	1	**4**	MCA	Theory+Lab	1.5	100+50	3	100+50
3.	MCA114AX	Professional Elective Course (Group-A)	3	0	1	**4**	MCA	Theory+Lab	1.5	100+50	3	100+50
4.	MCA115TB	AI Fundamentals	2	0	0	**2**	MCA	Theory	1.5	50	2	50
5.	MCA117IA	Data structures using Python	3	0	1	**4**	MCA	Theory+Lab	1.5	100+50	3	100+50
6.	HSS116EL	Technical English	0	0	1	**1**	HSS	Lab (Online)	1.5	50	1.5	50
7.	MMA001TA	*Bridge Course Mathematics	0	0	0	**0**	MA	Theory	1.5	50	-	-
Total Credits						**19**						

*** Bridge Course:** Non-Credit Mandatory Course MMA001TA -Bridge Course Mathematics: Students who have not taken Mathematics at the 10+2 or degree level are required to study and pass this course in the 1st semester. However, this course/subject will not be considered for vertical progression.

Note: 1 Credit = 30 hrs (L T P)

List of Professional Electives: I Semester

SL No	Course Code	Elective - A
1.	MCA114A1	Data Science
2.	MCA114A2	Computer Networks Concepts

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II SEMESTER MCA												
			Credit Allocation									
SL No	**Course Code**	**Course Title**	**L**	**T**	**P**	**Total Credits**	**BoS**	**Category**	**CIE Duration (H)**	**Max Marks CIE**	**SEE Duration (H)**	**Max Marks SEE**
1.	MCA221IA	Mobile Application Development	3	0	1	**4**	MCA	Theory+Lab	1.5	100+50	3	100+50
2.	MCA222IA	Design and Analysis of Algorithms	3	0	1	**4**	MCA	Theory+Lab	1.5	100+50	3	100+50
3.	MCA223TB	Agile Software Development	3	0	0	**3**	MCA	Theory	1.5	100	3	100
4.	MCA224IA	Object Oriented Programming using Java with Spring Boot	3	0	1	**4**	MCA	Theory+Lab	1.5	100+50	3	100+50
5.	MCA225CX	Professional Elective course (Group-C)	3	0	0	**3**	MCA	Theory	1.5	100	3	100
6.	MIM426RT	Research Methodology (MOOC)	2	0	0	**2**	IM	NPTEL	-	-	MOOC	100
7.	MCA427DI	Innovative Design Thinking lab	0	0	1	**1**	MCA	Lab	1.5	50	2	50
		Total Credits				**21**						

List of Professional Electives: II Semester

SL No	Course Code	Elective - C
1.	MCA225C1	Cyber Security
2.	MCA225C2	Gen-AI
3.	MCA225C3	Extended Reality
4.	MCA225C4	Software Testing

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III SEMESTER MCA												
			Credit Allocation									
SL No	**Course Code**	**Course Title**	**L**	**T**	**P**	**Total Credits**	**BoS**	**Category**	**CIE Duration (H)**	**Max Marks CIE**	**SEE Duration (H)**	**Max Marks SEE**
1.	MCA261IA	Full Stack Application Development	3	0	1	**4**	MCA	Theory+Lab	1.5	100+50	3	100+50
2.	MCA262IA	DevOps Automation	3	0	1	**4**	MCA	Theory+Lab	1.5	100+50	3	100+50
3.	MCA233TB	AI Product & Project Management	3	0	0	**3**	MCA	Theory	1.5	100	3	100
4.	MCA234EX	Professional Elective course (Group-E)	3	0	1	**4**	MCA	Theory+Lab	1.5	100+50	3	100+50
5.	MCA263DX	Professional Elective Course (Group-D)	3	0	0	**3**	MCA	Theory	1.5	100	3	100
6.	MCA461P	Minor Project	0	0	3	**3**	MCA	Lab	1.5	50	3	50
Total Credits						**21**						

List of Professional Electives: III Semester

SL No	Course Code	Elective - E	SL No	Course Code	Elective - D
1.	MCA234EA	Business Intelligence & Visualization Practices	1.	MCA263D2	Deep Learning
2.	MCA234EB	Internet of Things	2.	MCA263D3	Advanced Computer Networks
3.	MCA234EC	Cloud Computing Applications	3.	MCA263D4	Principles UI/UX Design
4.	MCA234ED	Agentic AI (Industry Elective)	4.	MCA263D5	E-Marketing

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IV SEMESTER MCA												
			Credit Allocation									
SL No	**Course Code**	**Course Title**	**L**	**T**	**P**	**Total Credits**	**BoS**	**Category**	**CIE Duration (H)**	**Max Marks CIE**	**SEE Duration (H)**	**Max Marks SEE**
1.	MCA241FX	Professional Elective Course (MOOC)	3	0	0	3		NPTEL	1.5	100	3	100
2.	MCA442P	Major Project	0	0	10	10	MCA	Lab	1.5	100	3	100
3.	MCA443N	* Project Internship	0	0	6	6	MCA	Lab	1.5	100	3	100
		Total Credits				**19**						

***To be undertaken after completion of 2nd semester and before commencement of 3rd semester (6 weeks duration)**

SEMESTER: I						
DISCRETE MATHEMATICS AND PROBABILITY THEORY **Category: Basic Science** **(Theory)**						
Course Code	:	**MMA211TD**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	:	**3:1:0**		**SEE**	**:**	**100 Marks**
Pedagogical Hours(L:T:P:SL)	**:**	**45:30:0:45**		**CIE duration**	**:**	**2.00 Hours**
Total credits	**:**	**4**		**SEE Duration**	**:**	**3.00 Hours**

UNIT – I	**9 Hours**
Graph Theory: Definition and examples of graphs, properties of a graph, sub graphs, regular graphs, bipartite graphs, paths and cycles, operations on graphs (union, intersection, Cartesian product), isomorphism of graphs. Eulerian graphs, Hamiltonian graphs, directed graphs, in degrees and out degrees in digraphs. Travelling salesman problem.	
UNIT – II	**9 Hours**
Logic: Basic connectivity and Truth table, Logical equivalence, logical implications, Quantifiers – Predicates: Predicative logic, Free and Bound variables, Rules of inference, Consistency. Proofs of theorems-direct, indirect, and proof by contradiction.	
UNIT – III	**9 Hours**
Elementary Linear Algebra: Rank of matrices-Rank of a matrix by Echelon form, consistency of system of linear equations- homogeneous and non-homogeneous equations, Gauss elimination, Gauss-Jordan and Gauss- Seidel methods. Eigenvalues and Eigenvectors-properties, largest eigenvalue and smallest eigenvalue by Rayleigh's power method	
UNIT – IV	**9 Hours**
Statistics and Probability: Curve fitting by method of least squares, fitting of curves – polynomial, exponential, power function. Correlation and linear regression analysis. Basic concepts of probability, conditional probability, Bayes' theorem.	
UNIT – V	**9 Hours**
Probability Distributions: Random variables- discrete and continuous, probability mass function, probability density function, and cumulative density function. Binomial distribution, Poisson distribution, Exponential distribution, and Normal distribution.	

Course Outcomes: After going through this course, the student will be able to:		
CO1	:	Formulate, analyze, and optimize structural and quantitative problems across graph structures, logical systems, linear algebraic systems, data distributions, and random variables.
CO2	:	Apply logical deduction, formal proof mechanisms, structural graph connectivity, algebraic transformations, and probabilistic models to validate system correctness and integrity.
CO3	:	Execute numerical procedures, matrix eliminations, linear algorithms, graph calculations, statistical curve fitting, and distribution parameters for computing tasks.
CO4	:	Analyze empirical datasets using statistical regression, correlation analysis, spectral matrix decomposition, path graph properties, and probabilistic evidence update models.

Reference Books	
1.	Ralph P Grimaldi, B. V. Ramana, Discrete and Combinatorial Mathematics, An applied Introduction, 5th Edition, Pearson Education, 2007, ISBN-10: 8177584243, ISBN-13:9788177584240..
2.	Kenneth H Rosen, Discrete Mathematics & its applications, 7th Edition, McGraw-Hill, 2010, ISBN-10: 0073383090, ISBN-13: 978-0-073383095.
3.	Probability & Statistics for Engineers & Scientists, Ronald E. Walpole & Raymond H. Myers, 9th Edition, 2016, Pearson Education, ISBN-13: 978-0134115856
4.	J.A. Bondy and U.S.R. Murty. Graph theory with Applications, Springer, 1st Edition, 2008.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**

SEMESTER: I						
WEB APPLICATION DEVELOPMENT **Category: Professional core** **(Theory + Practice)**						
Course Code	**:**	**MCA112IA**		**CIE**	**:**	**100 + 50 Marks**
Credits: L:T:P	**:**	**3:0:1**		**SEE**	**:**	**100 + 50 Marks**
Pedagogical Hours(L:T:P:SL)	**:**	**45:0:30:45**		**CIE duration**	**:**	**2.00 + 2.00 Hours**
Total credits	**:**	**4**		**SEE Duration**	**:**	**3.00+3.00 Hours**

UNIT – I	**9 Hours**

HTML5 and Modern Web Design: Introduction to Web Applications, Static and Dynamic Web Pages, Basic Web Application Architecture, HTML5 Document Structure, Semantic Elements, Text Formatting, Comments, Hyperlinks, Images, Lists, Tables, Multimedia Elements (Audio and Video), Forms and Form Controls.

CSS3: CSS Syntax and Selectors, Inline, Internal and External Style Sheets, Backgrounds, Fonts and Text Styling, Box Model, Margins, Padding, Borders, Positioning, Responsive Web Design, CSS Flexbox, CSS Grid Layout and Media Queries.

Web Design Principles: Introduction to User Interface (UI), User Experience (UX) and Accessibility Guidelines (WCAG).

UNIT – II	**9 Hours**

Bootstrap 5: Introduction, Grid System, Containers, Typography, Tables, Images, Buttons, Forms, Navigation Bar, Cards, Utility Classes and Responsive Components.

JavaScript: Introduction, Variables and Data Types, Operators and Expressions, Control Statements, Arrays, Functions, Document Object Model (DOM), Event Handling, ES6 Features (let, const and Arrow Functions), Fetch API, Regular Expressions (RegEx) for Form Validation.

Overview of Modern Front-end Development: Component-Based Development, Single Page Applications (SPA), Introduction to Front-end Frameworks and Libraries (React, Angular and Vue.js).

UNIT – III	**9 Hours**

Database Fundamentals and Conceptual Data Modelling

Introduction to Databases: Data, Information, Database, Database Management System (DBMS), Characteristics of the Database Approach, Data Models, Schema and Instances, Three-Schema Architecture and Data Independence, Database Languages and Interfaces, Client–Server Database Architecture.

Conceptual Data Modelling: Entity Types, Entity Sets, Attributes, Relationship Types, Relationship Sets, Roles, Structural Constraints, Weak Entity Types, ER Diagrams, Naming Conventions and Design Issues.

Introduction to Modern Database Systems: Relational, NoSQL and Vector Databases; Structured, Semi-Structured and Unstructured Data; Types of NoSQL Databases (Document, Key-Value, Column Family and Graph Databases); Embeddings, Similarity Search and Applications of Vector Databases in AI-enabled Web Applications.

UNIT – IV	**9 Hours**

Structured Query Language: Data Definition and Data Types, Specifying Constraints in SQL**.**

SQL Retrieval Queries: Basic Retrieval Queries in SQL, Nested Queries, EXISTS and UNIQUE, Join Tables and Outer Joins, Aggregate Functions, Views (Virtual Tables) and Specification of Views in SQL.

Data Manipulation: INSERT, DELETE and UPDATE Statements in SQL.
Transaction Management: Transaction Concept, ACID Properties, Basic Transaction Control Statements (COMMIT, ROLLBACK and SAVEPOINT).
Schema Modification: Schema Change Statements in SQL.

UNIT – V	**9 Hours**

Relational Model: Relational Model Concepts, Relational Model Constraints and Relational Database Schemas and Keys, Update Operations, Dealing with Constraint Violations, Relational Database Design Using ER-to-Relational Mapping
Normalization: Informal Design Guidelines for Relation Schemas, Functional Dependencies,
Normal Forms Based on Primary Keys, General Definitions of Second and Third Normal Forms, Boyce-Cod Normal form

LABORATORY

1. Design a responsive web portal using HTML5, CSS3 and Bootstrap 5 for an Online Book Store. The portal should consist of multiple web pages such as Home, About Us, Book Catalogue, User Profile and Contact Us. Incorporate semantic HTML5 elements, responsive layouts, navigation menus and Bootstrap components.
2. Develop an interactive web page for an Inventory Management System using HTML5, CSS3, Bootstrap 5 and JavaScript. Demonstrate the use of DOM Manipulation, Event Handling and Fetch API to dynamically update page content and improve user interaction.
3. Design a responsive Registration Form using HTML5 and Bootstrap 5. Validate user inputs using JavaScript and Regular Expressions (RegEx) for fields such as Name, Email, Password, Mobile Number and Date of Birth.
4. Design an Entity Relationship (ER) Diagram and implement the corresponding relational database for any one of the following domains:
    - Tourism
    - Human Resource Management
    - Debris Management
    - Library Management
    - Student Information System

   Create tables with appropriate constraints and relationships.
5. SQL Query Implementation - Implement SQL operations on the database created in Experiment 4 by performing:
    - Table Creation (DDL)
    - Data Insertion, Updation and Deletion (DML)
    - Retrieval Queries
    - Aggregate Functions
    - Joins
    - Nested Queries
    - Views
    - Transaction Control Statements (COMMIT, ROLLBACK and SAVEPOINT)

Course Outcomes:		
After going through this course, the student will be able to:		
CO1	:	Apply the concepts of modern web technologies and database systems for web application development.
CO2	:	Analyze web technologies and database solutions to meet different application requirements
CO3	:	Design web application architectures and database solutions for real-world applications
CO4	:	Develop responsive, database-driven web applications by integrating modern web technologies and database systems

Reference Books	
1.	Robert W. Sebesta, Programming the World Wide Web, Pearson Education, 10th Edition, 2018, ISBN: 9780133775983
2.	Jennifer Robbins, Learning Web Design: A Beginner's Guide to HTML, CSS, JavaScript, and Web Graphics, O'Reilly Media, 6th Edition, 2023. ISBN: 978-1098158996
3.	Marijn Haverbeke, Eloquent JavaScript, 4th Edition, No Starch Press, 2024. ISBN: 978-1718503403
4.	Ramez Elmasri, Shamkant B. Navathe, Fundamentals of Database Systems, Pearson, 7th Edition, 2017, ISBN 13: 978-9332582705

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
4.	**LAB:** Conduction of laboratory exercises, lab report, observation, and analysis (30 Marks), PBL progress review (10 marks) and lab test (10 Marks) adding up to 50 Marks. **THE FINAL MARKS WILL BE 50 MARKS**	**50**
	MAXIMUM MARKS FOR THE CIE	**150**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**

RUBRIC FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	**CONTENTS**	**MARKS**
Evaluation will be conducted through problem-/project-based learning activities specific to the program or laboratory. The assessment rubrics will be mapped to the respective course outcomes and shared with students along with the problem statements.		**50**

SEMESTER: I						
AI FUNDAMENTALS **Category: Professional core** **(Theory)**						
Course Code	:	**MCA115TB**		**CIE**	:	**50 Marks**
Credits: L:T:P	:	**2:0:0**		**SEE**	:	**50 Marks**
Pedagogical Hours(L:T:P:SL)	:	**30:0:0:30**		**CIE duration**	:	**2.00 Hours**
Total credits	:	**2**		**SEE Duration**	:	**2.00 Hours**

UNIT-I	10 Hrs
Introduction: What is AI? Acting humanly: The Turing test approach, Thinking humanly: The cognitive modelling approach, Thinking rationally: The "laws of thought" approach, **Acting rationally:** The rational agent approach; The foundations of AI: Mathematics, Economics, Neuroscience, Psychology, Computer Engineering; The State of the Art; Risks and Benefits of AI **Intelligent Agents:** Agents and Environments, The concept of Rationality, The Nature of Environments **Solving Problems by Searching:** Problem-solving agents: Search problems and solutions, Formulating problems, Example Problems: Grid world problems (Vacuum world, Sokoban Puzzle), Real-world problems (Route-finding problems)	
UNIT – II	**10 Hrs**
Search Algorithms: Uninformed Search Strategies (BFS, DFS); Informed (Heuristic) Search Strategies(A*Search) **Introduction to Machine learning:** Well-posed learning problems, designing a learning System: Choosing the training experience, Choosing the target function, choosing representation for the target function, Choosing a function approximation algorithm, the final design; Prospects and Issues in ML. **Instance-Based Learning:** K-Nearest Neighbor learning, and Applications	
UNIT –III	**10 Hrs**
Basic Neural Networks: Understanding the Biological neuron, Exploring Artificial Neuron, Types of Activation functions, Architectures of Neural network, Learning process in ANN. **Responsible AI and AI Governance**: Responsible AI, AI Governance, Accuracy - Risk Management Framework, Robustness - Measuring Robustness, Fairness and Human Rights - Measuring Fairness for Organizations, Measuring Fairness for Individuals, Privacy - Scoring Rubric, Sustainability - Environmental Sustainability, Social and Governance Sustainability, Human-Centered AI - Evaluating AI	

Course Outcomes: After completing the course, the students will be able to: -	
CO1	Apply the fundamental concepts of Artificial Intelligence, intelligent agents, problem-solving, machine learning, neural networks, and Responsible AI principles.
CO2	Analyze Artificial Intelligence techniques to solve computational problems.
CO3	Illustrate appropriate AI algorithms to design intelligent solutions for real-world applications while considering ethical and responsible AI principles.
CO4	Evaluate the performance, applicability, and societal impact of AI algorithms, for developing reliable and ethical intelligent systems.

Reference Books	
1.	Stuart J. Russell and Peter Norvig , Artificial Intelligence: A Modern Approach, Publisher: Pearson, 4th Edition, 2020, ISBN-13: 978 0134610993
2.	Tom M. Mitchell, Vineeth N Balasubramanian, Machine Learning, McGraw-Hill, 2nd Edition 2026, ISBN-13: 9789364446037
3.	Duke, Toju, and Paolo Giudici. *Responsible AI in Practice: A Practical Guide to Safe and Human AI*. Apress, 2025.
4.	Melanie Mitchell, Artificial Intelligence: A Guide for Thinking Humans, Publisher: Farrar, Straus and Giroux, 2019, ISBN-13: 978 0374257835

RUBRICS FOR THE CONTINUOUS INTERNAL EVALUATION (THEORY)		
Sl.N	**COMPONENTS**	**MARKS**
1.	**QUIZ:** One Quiz will be conducted in online/offline mode. & will be evaluated for 10 Marks. THIS WILL BE CONSIDERED AS THE FINAL QUIZ MARKS.	**05**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). **TWO TESTS** will be conducted. Each test will be evaluated for 25 Marks, adding up to 50 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 20 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**20**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (10) & Phase II (15) ADDING UPTO 25 MARKS**.	**25**
	MAXIMUM MARKS FOR THE CIE THEORY	**50**

RUBRICS FOR SEMESTER END EXAMINATION (THEORY)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	**16**
3 & 4	Unit 2: Question 3 or 4	**17**
5 & 6	Unit 3: Question 5 or 6	**17**
	TOTAL	**50**

SEMESTER: I						
DATA STRUCTURES USING PYTHON **Category: Professional core** **(Theory + Practice)**						
Course Code	**:**	**MCA117IA**		**CIE**	**:**	**100 + 50 Marks**
Credits: L:T:P	**:**	**3:0:1**		**SEE**	**:**	**100 + 50 Marks**
Pedagogical Hours(L:T:P:SL)	**:**	**45:0:30:45**		**CIE duration**	**:**	**2.00 + 2.00 Hours**
Total credits	**:**	**4**		**SEE Duration**	**:**	**3.00+3.00 Hours**

UNIT-I	**09 Hrs**
Python and Foundations of Data Structures and Algorithms — Introduction to Python: Python overview, objects in Python, identifiers, objects and assignment, expressions, operators and precedence, control flow and functions. Object-Oriented Programming: object-oriented design principles, class definitions and inheritance. Data Structures and Algorithms: elementary data organization, classification of data structures, operations on data structures, Abstract Data Types (ADT), experimental studies, growth of functions and analysis of algorithms using Big-Oh, Big-Omega and Big-Theta notations. Recursion: examples illustrating recursion, analyzing recursive algorithms, linear recursion, binary recursion and designing recursive algorithms.	
UNIT – II	**09 Hrs**
Arrays, Searching, Sorting, Stacks and Queues — Array-Based Sequences: Python sequence types, low-level arrays, referential arrays, dynamic arrays and implementation of a dynamic array, efficiency of Python sequence types. Searching and Sorting: Linear Search, Binary Search using recursion, Bubble-Sort and Selection-Sort. Stacks: Stack ADT, array-based stack implementation, stack operations, matching parentheses and evaluation of arithmetic expressions. Queues: Queue ADT, array-based queue implementation, queue operations, circular queues, using an array circularly and implementation of a circular queue using an array.	
UNIT –III	**09 Hrs**
Linked Lists, Trees and Binary Search Trees - Linked Lists: singly linked lists, circularly linked lists, doubly linked lists and link-based versus array-based sequences. Trees: Tree ADT, binary trees, tree representations and implementations, preorder, postorder, inorder and breadth-first traversals. Binary Search Trees: binary search tree organization, searching, insertion and deletion operations, Python implementation and performance of binary search trees.	
UNIT –IV	**09 Hrs**
Hash Tables, Heaps and Priority Queues — Maps and Hash Tables: Map ADT, hash-table implementation, hash functions, collision-handling schemes, separate chaining, open addressing, load factors and rehashing. Priority Queues: Priority Queue ADT and priority queue implementations. Heaps: heap data structure, array-based heap representation, heap operations, heap-based priority queue implementation and analysis.	
UNIT –V	**09 Hrs**
Graphs and Graph Traversals — Graphs: graph terminology and properties, Graph ADT, graph representations using edge list, adjacency list, adjacency map and adjacency matrix. Graph Traversals: Depth-First Search (DFS), DFS implementation and extensions, Breadth-First Search (BFS).	

LABORATORY
1. Array ADT and Searching: Implement an Array ADT using a fixed-capacity Python list as the underlying storage. Perform traversal, insertion, deletion, updating and searching operations. Implement Linear Search and Binary Search on an ordered array without using Python built-in methods that directly perform the required operations. 2. Sorting and Binary Search: Implement Bubble-Sort and Selection-Sort without using built-in sorting functions. Implement Binary Search using recursion and compare the number of comparisons/operations for different input sizes. 3. Stack ADT and Applications: Implement a Stack ADT using a fixed-capacity Python list with an explicit top. Implement push, pop, peek, isEmpty and isFull operations. Use the stack for parenthesis matching and postfix expression evaluation. 4. Linear and Circular Queues: Implement Linear Queue and Circular Queue using fixed-capacity Python lists with explicit front and rear. Implement enqueue, dequeue and peek operations, including overflow and underflow conditions. 5. Singly Linked List: Implement a Singly Linked List using a user-defined Node class and references. Perform insertion at the beginning, end and specified position; deletion, searching, traversal and reversal. 6. Doubly and Circularly Linked Lists: Implement Doubly Linked List and Circularly Linked List using node objects and references. Perform insertion and deletion operations and appropriate traversals, including forward and backward traversal of a Doubly Linked List. 7. Binary Trees and Binary Search Trees: Represent a Binary Tree using node objects. Implement recursive preorder, inorder and postorder traversals and breadth-first traversal. Implement Binary Search Tree operations for insertion, searching and deletion. 8. Hash Table: Implement a fixed-size Hash Table using a suitable hash function. Perform insertion, searching and deletion using separate chaining. Demonstrate an open-addressing collision-resolution method such as linear probing. 9. Heap and Priority Queue: Implement a Binary Min-Heap using an array representation. Perform insertion, deletion/extract-min and heapify operations. Extend the heap implementation to realize a Priority Queue and demonstrate the relationship between min-heaps and max-heaps. 10. Graph Representation and Traversals: Represent a graph using an adjacency-list representation and implement Breadth-First Search (BFS) and Depth-First Search (DFS). Use a Queue for BFS and recursion or an explicit Stack for DFS

Course Outcomes	
CO1	Apply fundamental concepts of Python, abstract data types, data structures, and algorithms to model, organize, and manipulate data for problem-solving contexts.
CO2	Implement appropriate data structures and algorithms using Python to perform fundamental data processing operations
CO3	Analyze the characteristics, representations, and operations of linear and non-linear data structures through suitable examples and implementations.
CO4	Evaluate the suitability of data structures and algorithms for solving data organization and processing problems based on application requirements

References	
1.	M. T. Goodrich, R. Tamassia, and M. H. Goldwasser, Data Structures and Algorithms in Python, Wiley India, 2021, ISBN: 978-9354247866.
2.	John Canning, Alan Broder and Robert Lafore, Data Structures & Algorithms in Python, Pearson Education, 2023, ISBN-13: 978-0134855684
3.	Benjamin Baka, Python Data Structures and Algorithms, Packt Publishing, Second Edition, 2023. ISBN-13: 978-1801073448
4.	Kent D. Lee and Steve Hubbard, Data Structures and Algorithms with Python, Springer International Publishing, First Edition, 2015.

BRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
4.	**LAB:** Conduction of laboratory exercises, lab report, observation, and analysis (30 Marks), PBL progress review (10 marks) and lab test (10 Marks) adding up to 50 Marks. **THE FINAL MARKS WILL BE 50 MARKS**	**50**
	MAXIMUM MARKS FOR THE CIE	**150**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**
RUBRIC FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	**CONTENTS**	**MARKS**
Evaluation will be conducted through problem-/project-based learning activities specific to the program or laboratory. The assessment rubrics will be mapped to the respective course outcomes and shared with students along with the problem statements.		**50**

SEMESTER: I						
DATA SCIENCE **Category: Professional Elective (Group A)** **(Theory + Practice)**						
Course Code	:	**MCA114A1**		**CIE**	:	**100 + 50 Marks**
Credits: L:T:P	:	**3:0:1**		**SEE**	:	**100 + 50 Marks**
Pedagogical Hours(L:T:P:SL)	:	**45:0:30:45**		**CIE duration**	:	**2.00 + 2.00 Hours**
Total credits	:	**4**		**SEE Duration**	:	**3.00+3.00 Hours**

UNIT - I	**9 Hours**
Introduction to Data Science: Data Science, Brief History of Data Science, increasing attention to data science, Fundamental fields of study to data science, Data science and Related Terminologies, Types of Analytics, Application of Data Science, Data Science Process Model, **Python environment and basics of Python:** Jupyter notebook, setting working directory in python, variables, data types, operators, functions in python	
UNIT - II	**9 Hours**
Python for Data Science – Python libraries, Numpy Library, Pandas Library; **Preparing to model**: Basic types of data, Exploring structure of data, Data quality and remediation, Data pre-processing: Dimensionality Reduction, Feature subset selection, Independent and Dependent Variables, Data visualization using Python-Matplotlib Library, Seaborn Library	
UNIT - III	**9 Hours**
Machine Learning and Supervised Learning Models: Types of Machine learning algorithms, Supervised and Unsupervised Learning Algorithms, Selecting a model-predictive and descriptive models, Training a model for supervised learning- holdout method, k-fold cross validation method, Bootstrapping, Lazy vs Eager Learning, Overfitting and under fitting, correctness, The bias-variance tradeoff, Feature Transformation, Feature subset selection.	
UNIT - IV	**9 Hours**
Supervised Learning Algorithms: Model performance measures, Naïve Bayes algorithm: Poisson distribution, binomial and multinomial models, Linear Regression, Building linear regression, Interpretation of Linear Regression coefficients, Validation of Linear regression, Decision Tree, Tree Structure, Criteria for splitting decision node, Random Forest model	
UNIT - V	**9 Hours**
Unsupervised Learning: Introduction, applications of Unsupervised learning, Clustering, Distance Measures, Distance Matrix, Linkage Methods, K Means clustering, K-Medoids clustering, Evaluation of Clusters, finding pattern using Association Rule Mining, **Text Analytics**: Unstructured data, Word Cloud **Introduction to Quantum Computing:** Fundamentals of Quantum Computing, Classical Computing vs. Quantum Computing, Qubits and Quantum States, Superposition and Quantum Entanglement, Quantum Gates and Circuits, Overview of Grover's Search, Shor's Algorithm, Applications of Quantum Computing	

LABORATORY

1. Implement NumPy and pandas' operations for Data Science concept(five each)
2. Consider the automobile dataset and perform exploratory data analysis.
   - Identify the dimension, structure, and summary of the data set
   - Preprocess the dataset and treat them (like missing values, 'na', ?). Justify the treatment
   - Plot the histogram for continuous variables (at least two) to analyze the data.
   - Draw a violin plot do describe the distribution of a numerical variable to analyze the data.
   - Recognize the outliers using box plot (Display the box plot before and after outlier treatment)
   - Display a heat map to display the relationship among the attributes
   - Standardize the continuous variable (if any)
3. For the data set in Q2,
   - Show the distribution of continuous variables using histogram
   - Identify the relationship between two continuous variables using scatter plot
   - Find and display the frequency of the categorical values using count plot
   - Apply point plots to display one continuous and one categorical variable
4. Consider the health care dataset that consists of several imaging details from patients that had a biopsy to test for breast cancer. The variable diagnosis classifies the biopsied tissue as M = malignant or B = benign. Describe and pre-process the dataset. Use KNN supervised learning model to predict Diagnosis using texture mean and radius mean . Analyze the model using different k values and display the performance of the model
5. Consider the student performance dataset. Predict the student performance as "Pass" or "Fail" by implementing a decision tree. Perform data preprocessing and visualize the data. Identify the important features affecting the student performance and Analyze the efficiency of the decision tree using different metrics. Plot the decision tree.
6. For the dataset in Q5, apply random forest algorithm to predict the student performance.
   - Plot the important variables using seaborn
   - Tune the random forest for training and test data based on best parameters and implement it
   - Analyze the model performance and display the output
7. For the placement dataset apply Naïve bayes classification
   - Plot the prior probability and posterior probability
   - Build the Naïve Bayes classifier
   - Analyze the model performance
8. For the market basket dataset, apply apriori algorithm and identify the best rules based on Support and confidence values.
9. For the Mall-Customers dataset Implement k-means clustering algorithm and visualize the Clusters
10. Create different word cloud for the provided text file.

Course Outcomes: After going through this course the student will be able to:		
CO1	:	Apply data science methodologies, Python programming, data preprocessing, visualization techniques, and appropriate analytical methods to solve data-driven problems.
CO2	:	Analyze datasets using data science methodologies, feature engineering, machine learning techniques, and validation methods to develop suitable analytical models.
CO3	:	Develop predictive and descriptive data science models using appropriate machine learning algorithms, clustering techniques, text analytics methods, for extracting meaningful insights from data.
CO4	:	Asses the performance, applicability of data science solutions and emerging quantum computing approaches for intelligent decision-making in real-world applications.

Reference Books	
1.	B Uma Maheswari, R Sujatha, Introduction to Data Science Practical Approach with R and Python, Wiley Publications, 1st Edition, 2021, ISBN-: 9789354640506, ISBN-13: 9789354640513 (EBook)
2.	Saikat Dutt, Subramanian Chandramouli, Amith Kumar Das, Machine Learning, Pearson publications, 2nd edition, 2025, ISBN978-81-198-9673-8.
3.	Laura Igual, Santi Seguí, Springer Publications, Introduction to Data Science- A Python Approach to Concepts, Techniques and Applications, 2024, ISSN: 1863-7310 ISSN 2197-1781 (electronic)
4.	Joel Grus, Data Science from Scratch, First principles with Python, O'Reilly, 2nd Edition, 2019, ISBN: 9781492041108, 1492041106
5.	https://archive.nptel.ac.in/courses/106/106/106106212/ https://www.nptel.ac.in/courses/106106232?

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
4.	**LAB:** Conduction of laboratory exercises, lab report, observation, and analysis (30 Marks), PBL progress review (10 marks) and lab test (10 Marks) adding up to 50 Marks. **THE FINAL MARKS WILL BE 50 MARKS**	**50**
	MAXIMUM MARKS FOR THE CIE	**150**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**

RUBRIC FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	**CONTENTS**	**MARKS**
Evaluation will be conducted through problem-/project-based learning activities specific to the program or laboratory. The assessment rubrics will be mapped to the respective course outcomes and shared with students along with the problem statements.		**50**

SEMESTER: I						
COMPUTER NETWORKS CONCEPTS **Category: Professional Elective (Group A)** **(Theory + Practice)**						
Course Code	**:**	**MCA114A2**		**CIE**	:	**100 + 50 Marks**
Credits: L:T:P	**:**	**3:0:1**		**SEE**	:	**100 + 50 Marks**
Pedagogical Hours(L:T:P:SL)	**:**	**45:0:30:45**		**CIE duration**	:	**2.00 + 2.00 Hours**
Total credits	**:**	**4**		**SEE Duration**		**3.00+3.00 Hours**

UNIT – I	**9 Hours**
The Unix/Unix Like Operating System architecture and commands: Unix Architecture, General purpose networking commands (ping, ifconfig, ARP, SSH) Introduction: Introduction, Network Hardware, Network Software: Protocol Hierarchies, Design Issues for the Layers **Reference Models:** The OSI Reference Model, The TCP/IP Reference Model, A Comparison of the OSI and TCP/IP Reference Models, Physical Layer-Guided Transmission Media	
UNIT – II	**9 Hours**
Data Link Layer: Data link Layer Design issues, Error Detection codes, Sliding Window Protocols (Stop and Wait, Go-Back-N (GBN) and Selective Repetitive (SR)) **Medium Access Control:** The Channel Allocation Problem, Multiple Access Protocols, Ethernet	
UNIT – III	**9 Hours**
The Network Layer: Network Layer Design issues, Routing algorithms- The Optimality Principal, Fooding, Distance Vector Routing, Link State Routing, Hierarchical routing, Congestion Control Algorithms. The Network Layer in the internet- IP version 4 Protocol, **IP version 6 protocol:** The Main IPv6 Header, Extension Headers, Internet Control Protocols: ICMP, ARP, DHCP	
UNIT – IV	**9 Hours**
The Transport Layer: The Transport Service: Services Provided to the Upper Layers, Transport service primitives, Berkeley Sockets, Elements of Transport Protocols, **Internet transport protocols** - TCP: TCP service model, TCP protocol, Segment Header, Connection establishment, connection release, TCP sliding window, UDP, RPC	
UNIT – V	**9 Hours**
The Application Layer: The Domain Name System, Name Servers, Electronic Mail: Architecture and Service, MIME, SMTP, POP, The World-Wide-Web, Streaming Audio and Video	

LABORATORY	**30 Hours**
1. Create a LAN with three or more nodes implementing star topology and demonstrate classful 2. Demonstrate secured file transfer and computing over wired network or a wireless network with SCP and SSH key based computing 3. Create a LAN using physical networks and install FTP server to demonstrate file transfer 4. Configure routing between two subnets using static routes and dynamic routing protocols (RIP/OSPF) and verify connectivity. 5. Build DHCP server using dns-masq with and without MAC binding with IPV4 and IPV6 6. Build DNS server for resolving the names and IP addresses 7. Configure a Network File System (NFS) server and client on a physical LAN to enable secure file sharing and access between Linux systems. 8. Deploy and Test a Mail Transfer Service on a physical LAN 9. Demonstrate multiple client server communication on different ports using netcat utility tool 10. Demonstrate Proxy - Server setup for a web server and SSH port forwarding	

Course Outcomes	
CO1	Apply the concept of layered approach in designing computer networks in real time
CO2	Analyze the design issues, services, interfaces and protocols for data flow in computer networks
CO3	Demonstrate the protocols and services designed for the layered approach
CO4	Evaluate the principles and protocols of computer networks for real time

References	
1.	Andrew S. Tanenbaum, David J Wetherall, Computer Networks, Pearson Publication, 6th Edition, 2021, ISBN-13: 9780137523214
2.	Behrouz A Forouzan, Firouz Mosharraf, Computer Networks A Top-Down Approach, McGraw-Hill, 1st Edition, 2023, ISBN 13: 9789355324900
3.	Sumitabha Das, Unix: Concepts and Applications, McGraw-Hill, 4th Edition, 2017, ISBN 13:978-0070635463
4.	Stallings, William. Data and computer communications. Pearson Education India, 9th Edition, 2013, ISBN13: 978-9332518865

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
4.	**LAB:** Conduction of laboratory exercises, lab report, observation, and analysis (30 Marks), PBL progress review (10 marks) and lab test (10 Marks) adding up to 50 Marks. **THE FINAL MARKS WILL BE 50 MARKS**	**50**
	MAXIMUM MARKS FOR THE CIE	**150**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**
RUBRIC FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	**CONTENTS**	**MARKS**
Evaluation will be conducted through problem-/project-based learning activities specific to the program or laboratory. The assessment rubrics will be mapped to the respective course outcomes and shared with students along with the problem statements.		**50**

SEMESTER: I						
TECHNICAL ENGLISH **Category: Online English Laboratory Course** **(Practice)**						
Course Code	:	**HSS116EL**		**CIE**	:	**50 Marks**
Credits: L:T:P	:	**0-0-1**		**SEE**	:	**50 Marks**
Pedagogical Hours(L:T:P:SL)	:	**0:0:30:0**		**CIE duration**	:	**1 Hours**
Hours	:	**30L**		**SEE Duration**		**2 Hours**
UNIT – I						**10 Hours**
The Basics. Business Documents, Questions, and the Technical Pursuit. Engineering Concepts and Complexity; The Future Tense for Technical Work. White Papers; Modifiers and Qualifiers.						
UNIT – II						**10 Hours**
Making Recommendations; Interpreting Data, Ethical Persuasion for Technical Projects; Cause and Effect; Calls for Proposals. Technical Complexity in Communication. Numbers, Plain English, Jargon, and Technical Terms, Active and Passive Structures						
UNIT – III						**10 Hours**
Organization Needs; Seeing the Big Picture; Negotiating. Audience Needs and Assessment; Standards versus White Papers; Objectivity, communicating within Expected Genres; Identifying Trustworthy Sources or Bias in. A Review of Major Course Takeaways						

Course Outcomes: **After going through this course, the student will be able to**:		
CO1	:	Demonstrate clarity and precision in technical communication by structuring information effectively, balancing technical terms with plain English, and adapting to diverse audiences.
CO2	:	Analyze and produce professional documents, such as white papers, business proposals, and reports, while applying ethical persuasion, data interpretation, and evidence-based reasoning.
CO3	:	Evaluate and refine communication strategies by assessing audience needs, recognizing trustworthy sources, and navigating organizational and technical complexities.
CO4	:	Apply critical thinking and negotiation skills to align communication with organizational goals, anticipate future challenges, and support informed decision-making.

References
1. IEEE – EBSCO Technical English for Professionals – Online platform
2. Valerie Lambert, Elaine Murray, English for Work – Everyday Technical English, Pearson Education, 2003, ISBN- 0 582 53963 3
3. David Bonamy, Christpher Jacques, Technical English – First Course Book, Pearson Education, 2008
4. S Sumant. Technical English I, The McGraw Hill, 2011, ISBN -978 81 8209 308 9

Assessment and Evaluation Pattern (Online Mode)		
	CIE (Online Mode)	**SEE (Online Mode)**
Weightage	**50%**	**50%**
Modules Assessment	**30 marks**	
Test – I	Each test will be conducted for 10 marks adding to 20 marks.	**Final assessment will be conducted for 50 marks**
Test – II		
Maximum Marks	50 Marks	**50 Marks**
Total marks for the course	**50**	**50**

SEMESTER: I						
BRIDGE COURSE MATHEMATICS **Category: Audit Course** **(Theory)**						
Course Code	:	**MMA001TA**		**CIE**	:	**50 Marks**
Credits: L:T:P	:	**0-0-0**		**SEE**	:	**NO SEE**
Pedagogical Hours(L:T:P:SL)	:	**30L**		**CIE duration**	:	**1.5 Hours**
Total credits		**00**		**SEE Duration**		-

UNIT - I	**6 Hours**
Set Theory: Basics of set theory, Cartesian product of sets, Relations, Properties of relations, Equivalence relations. Function composition and Inverse function (basic problems).	
UNIT - II	**6 Hours**
Sequence and Series: Sequences, Series, Arithmetic Progression (AP), Sum of Finite number of terms in AP, Arithmetic Means (AM), Geometric Progression (GP), sum to n terms of GP, Geometric Mean (GM), relation between AM and GM(basic problems.	
UNIT - III	**6 Hours**
Matrices and Determinants: Matrices, Types of matrices, Scalar multiplication, Addition of matrices, Product of Matrices Transpose of a matrix, Determinant of a matrix, Singular matrix and Inverse of a matrix (basic problems).	
UNIT - IV	**6 Hours**
Calculus: Limits- properties of limits, limits of Trigonometric Functions, Continuity (basic problems). **Derivatives:** Algebra of derivative of functions, polynomial, trigonometric function, product rule, quotient rule (basic problems). **Integrals:** Properties of integrals, Fundamental Theorem of Calculus (basic problems).	
UNIT - V	**6 Hours**
Statistics: Basic statistical concepts, qualitative and quantitative data, Classification of data, Construction of frequency distribution. Measure of central tendency—mean, median and mode. Measures of dispersion—standard deviation (basic problems)	

Course Outcomes:

After going through this course, the student will be able to:

CO1	:	Explore the fundamental concepts of Set theory, Sequence and Series, Matrices and Determinants, Calculus and Statistics.
CO2	:	Apply theoretical concept of Set theory, Sequence and Series, Matrices and Determinants, Calculus and Statistics to Formulate the problems in engineering problem.
CO3	:	Analyze the solution of the modern engineering problems solved using appropriate techniques of Set theory, Sequence and Series, Matrices and Determinants, Calculus and Statistics to optimize the solutions of engineering problem.
CO4	:	Develop the overall mathematical knowledge gained to demonstrate and analyze the Problems arising in real world situations.

Reference Books
1. Ralph P Grimaldi, B.V.Ramana, Discrete and Combinatorial Mathematics, An applied Introduction, 5th Edition, Pearson Education, 2007, ISBN-10: 8177584243, ISBN-13:9788177584240.
2. Kenneth H Rosen, Discrete Mathematics & its applications, 7th Edition, McGraw-Hill, 2010, ISBN-10: 0073383090, ISBN-13: 978-0-073383095.
3. B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers,44th Edition, 2015, ISBN: 978-81-933284-9-1.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quizzes (Q) and TWO Tests (T) component **[10 (Q) + 40 (T) = 50 marks)**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** One Quiz will be conducted in online/offline mode. & will be evaluated for 10 Marks. THIS WILL BE CONSIDERED AS THE FINAL QUIZ MARKS.	**10**
2.	**TESTS:** Students will be evaluated in test, descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analyzing, Evaluating, and Creating). **TWO TESTS** will be conducted. Each test will be evaluated for 25 Marks, adding up to 50 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.**	**40**
	MAXIMUM MARKS FOR THE CIE	**50**

SEMESTER: II						
OBJECT ORIENTED PROGRAMMING USING JAVA WITH SPRING BOOT **Category: Professional core** **(Theory + Practice)**						
Course Code	:	**MCA224IA**		**CIE**	:	**100 + 50 Marks**
Credits: L:T:P	:	**3:0:1**		**SEE**	:	**100 + 50 Marks**
Pedagogical Hours(L:T:P:SL)	:	**45:0:30:45**		**CIE duration**	:	**2.00 + 2.00 Hours**
Total credits	:	**4**		**SEE Duration**	:	**3.00+3.00 Hours**

UNIT – I	**9 Hours**
Object Oriented Design: Introduction to Object-Oriented Concepts, How to Think in Terms of Objects, The Anatomy of a Class, Class Design Guidelines. **Designing with Objects Inheritance**: Mastering Inheritance and Composition, Designing with Interfaces and Abstract Classes, Building Objects and Object-Oriented Design, categories of Design Patterns, The SOLID Principles of Object-Oriented Design.	
UNIT – II	**9 Hours**
Java Programming Fundamentals: Introduction to Java Programming, Applications of Java Programming, The Java Development Kit, The Java Keywords, Identifiers in Java, User input using Scanners, The Scope and Lifetime of Variables, operators, Conditional and Control Statements, Arrays, String Handling, **Objects and Methods:** How Objects are Created, Reference Variables and Assignment, Methods, Returning from a Method, Returning Value, Using Parameters, Constructors, this Keyword, Understanding Static.	
UNIT - III	**9 Hours**
Inheritance, Interface, Packages: Inheritance: Fundamentals, Overloading, Overriding, super keyword, Using final. Interface: Fundamentals, Creating an Interface, Implementing an Interface, Using Interface References, Implementing Multiple Interfaces. Package**:** Fundamentals, Importing Packages, Creating user defined packages	
UNIT - IV	**9 Hours**
Exception Handling: Exception Handling: Fundamentals, Hierarchy, Catching subclass Exceptions, try blocks can be nested, Throwing an Exception, Throwable, using finally, Java's Builtin Exceptions **Threads**: The Thread Class and Runnable Interface, Creating Thread, Creating Multiple Threads, Thread Priorities, Synchronization, using Synchronization Methods, Thread Communication using notify (), wait () and notify All(), Executor Framework, Thread Pools, Callable and Future	
UNIT - V	**9 Hours**
Collections framework: Collections Overview, The collection Classes and interfaces **Generics:** Generics Fundamentals, Generic Classes, Generic Methods, Bounded Types, Wildcards and Generic Restrictions. **Spring Boot Fundamentals:** Introduction to Spring Framework and Spring Boot, Spring Boot Architecture, creating a Project with Spring Boot Initializer, Dependency Injection and Inversion of Control (IoC), Spring Beans, creating a Simple Spring Boot Application; **Spring Boot RestAPI:** Writing RESTful controllers, Consuming REST services; Handling GET, POST, PUT and DELETE Requests	

LABORATORY

Write Java programs to demonstrate the following concepts. For each experiment, students shall use AI-assisted development tools (such as ChatGPT, GitHub Copilot, Gemini, Claude, etc.) to generate, review, optimize, debug, document and test the code, and submit a brief AI usage report.

1. a. scope of variable b. this keyword
2. a. constructor overloading and method overloading b. static keyword
3. The concepts of packages
4. a. Abstraction b. Run Time Polymorphism
5. User defined Exception Handling using throw, throws
6. Thread Life Cycle
7. Write a Java Program to demonstrate Generic classes and methods
8. Producer Consumer design pattern using Thread wait and notify methods
9. Demonstrate Dependency Injection using constructor based using Spring boot
10. Develop a Spring Boot RESTful Web Application to demonstrate the implementation of CRUD operations using HTTP methods GET, POST, PUT, and DELETE. Design appropriate REST APIs, validate inputs, and test the endpoints using AI-assisted API testing tool.(Open ended Lab Experiment)

Course Outcomes:

After going through this course, the students will be able to:

CO1	:	Apply object-oriented principles to design and implement real-world Java applications.
CO2	:	Design Java-based solutions to achieve modular and reusable code structures.
CO3	:	Analyze real-world problems and implement robust solutions using Java concepts.
CO4	:	Develop robust, efficient, and scalable software solutions by integrating appropriate Java programming concepts and techniques.

Reference Books

1. Matt Weisfeld, Object-Oriented Thought Process, Addison-Wesley Professional, 5th Edition, 2019, ISBN: 9780135182130
2. Gazihan Alankus, Rogério Theodoro de Brito, Basheer Ahamed Fazal, Vinicius Isola, Miles Obare, Java Fundamentals, Packt Publishing; 1st Edition (15 March 2019), ISBN-13:978- 1789801736
3. Herbert Schildt, Dale Skrien, Java Fundamentals, A Comprehensive Introduction , Tata McGraw Hill, 1st Edition, 2017,ISBN-13:978-1259006593
4. Jeff Friesen, Java Threads and the Concurrency Utilities, 1st Edition, Apress,ISBN-13:978-1484216996
1. K. Siva Prasad Reddy, Sai Upadhyayula, Beginning Spring Boot 3: Build Dynamic CloudNative Java Applications and Microservices, Apress Publications, November 2022, ISBN: 9781484287927

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
4.	**LAB:** Conduction of laboratory exercises, lab report, observation, and analysis (30 Marks), PBL progress review (10 marks) and lab test (10 Marks) adding up to 50 Marks. **THE FINAL MARKS WILL BE 50 MARKS**	**50**
	MAXIMUM MARKS FOR THE CIE	**150**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**
RUBRIC FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	**CONTENTS**	**MARKS**
Evaluation will be conducted through problem-/project-based learning activities specific to the program or laboratory. The assessment rubrics will be mapped to the respective course outcomes and shared with students along with the problem statements.		**50**

SEMESTER: II						
MOBILE APPLICATION DEVELOPMENT **Category: Professional core** **(Theory + Practice)**						
Course Code	:	**MCA221IA**		**CIE**	:	**100 + 50 Marks**
Credits: L:T:P	:	**3:0:1**		**SEE**	:	**100 + 50 Marks**
Pedagogical Hours(L:T:P:SL)	:	**45:0:30:45**		**CIE duration**	:	**2.00 + 2.00 Hours**
Total credits	:	**4**		**SEE Duration**	:	**3.00+3.00 Hours**

UNIT – I	**9 Hours**
Introduction to PWAs and Tooling: Intro. to Progressive Web Apps, Tools to Measure Progressive, WebApps. **PWA Features - Service Workers**: Promises, Fetch, Service Worker, Register the Service Worker, Updating, Service, Worker. Caching and Offline Functionality with Service Workers: The Fetch API, Cache API, going Offline, Different Caching Strategies.	
UNIT – II	**9 Hours**
Features to Use: Adding your App to the Home Screen with Web App Manifest, Turning a Real App into a PWA **Notifications:** Web Notifications: Requesting Permission to Notify, sending a Notification, Tagging Notifications, Web Notifications with Service Workers.	
UNIT - III	**9 Hours**
Learning Dart Basics - Why Use Dart? Commenting Code, Running the main() Entry Point, Referencing Variables, Using Operators, Declaring Variables, Using Flow Statements, Using Functions, Import Packages, Using Classes, Implementing Asynchronous Programming	
UNIT - IV	**9 Hours**
Introducing Flutter - Defining Widgets and Elements **Installing the Flutter SDK** - Installing on Linux, System Requirements, Get the Flutter SDK, Check for Dependencies **Understanding Widget Lifecycle Events** - The Stateless Widget Lifecycle, The Stateful Widget Lifecycle	
UNIT - V	**9 Hours**
Laying Out Your Widgets - Laying out the whole scene, MaterialApp widget The Scaffold widget, The AppBar widget **Using Basic Widgets -** Text, Container, Column, Row, Button **Custom gestures for your custom widgets** - Decide on your gestures and behaviors, Create your custom widget, Add a Gesture Detector widget, Associate your gesture with its behavior **Navigation and Routing** - Stack navigation, Drawer navigation, Tab Navigation	

LABORATORY

1. Build a simple web page using PWA by adding a Service Worker
2. Create a login page to authenticate a user using PWA with Manifest file
3. Demonstrate online and offline web page load using PWA, Service Worker and Caching
4. Build an application to do a stock display using PWA using a raw JSON file
5. Code a Click Counter App in Flutter using a basic UI layout with state management
6. Create a Message Toggle Application using the basics widgets in Flutter
7. Demonstrate the use the Gesture Detector widget to respond to double-tap and swipe events using Flutter
8. Build a simple counter app with increment and decrement buttons to show subsequent values on screen
9. Develop a Smart User Registration Form using Flutter widgets with TextField, Button, and input validation to provide intelligent password-strength analysis and username suggestions.
10. Design an Intelligent Theme Switcher Application using a Stateful Widget to automatically recommend Light or Dark mode based on the time of day or user-selected usage scenario.

Course Outcomes:

After going through this course, the students will be able to:

CO1	:	Apply core Android development concepts by creating activities, intents, and layouts, and managing threads for efficient data handling and UI updates.
CO2	:	Asses caching strategies and offline functionality using service workers and the Fetch API for Progressive Web Applications (PWAs).
CO3	:	Design user-engaging web apps by implementing push notifications, web manifests, and custom notification behaviors with service workers.
CO4	:	Construct cross-platform mobile applications using Flutter with interactive UI widgets, Dart programming principles and managing stateful & stateless widget lifecycles.

Reference Books

1. Dennis Sheppard, Beginning Progressive Web App Development: Creating a Native App Experience on the Web, Apress Publishing, 2017, ISBN: 9781484230909
2. Rap Payne, Beginning App Development with Flutter, Apress Publishing, 2019, ISBN: 9781484251805
3. Marco L. Napoli, Beginning Flutter: A Hands-On Guide to App Development, Wiley Publications, 2019, ISBN: 9781119550822
4. Frank Zammetti, Practical Flutter: Improve your Mobile Development with Google's Latest Open-Source SDK, 2019, ISBN: 978-1-4842-4971-0

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
4.	**LAB:** Conduction of laboratory exercises, lab report, observation, and analysis (30 Marks), PBL progress review (10 marks) and lab test (10 Marks) adding up to 50 Marks. **THE FINAL MARKS WILL BE 50 MARKS**	**50**
	MAXIMUM MARKS FOR THE CIE	**150**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**
RUBRIC FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	**CONTENTS**	**MARKS**
Evaluation will be conducted through problem-/project-based learning activities specific to the program or laboratory. The assessment rubrics will be mapped to the respective course outcomes and shared with students along with the problem statements.		**50**

SEMESTER: II						
DESIGN AND ANALYSIS OF ALGORITHM **Category: Professional core** **(Theory + Practice)**						
Course Code	:	**MCA222IA**		**CIE**	**:**	**100 + 50 Marks**
Credits: L:T:P	:	**3:0:1**		**SEE**	**:**	**100 + 50 Marks**
Pedagogical Hours(L:T:P:SL)	:	**45:0:30:45**		**CIE duration**	**:**	**2.00 + 2.00 Hours**
Total credits	:	**4**		**SEE Duration**	**:**	**3.00+3.00 Hours**

UNIT – I	**9 Hours**
Introduction to fundamentals of Algorithms in Computing: Algorithms Definition, characteristics, Design and analysis methodology, Problem types, Pseudocode. **Fundamentals of the Analysis of Algorithm Efficiency:** The Analysis Framework, Measuring an Input's Size, Units for Measuring Running Time, Orders of Growth, Worst-Case, Best-Case, and Average-Case Efficiencies, Asymptotic Notations and Basic Efficiency Classes, formal definitions of *Big-Oh, Big-Omega and Big- Theta* notation, Basic Efficiency Classes.Mathematical Analysis of Non-recursive Algorithms, Mathematical Analysis of Recursive Algorithms.	
UNIT – II	**9 Hours**
Decrease and conquer: Insertion Sort, Topological order **Divide and Conquer:** Merge sort and Quick Sort with performance measurement **Sorting and Order Statistics:** Sorting in linear time (Radix and Bucket Sort), Heap Sort	
UNIT – III	**9 Hours**
Greedy Method: Representation of Graphs, Knapsack Problem, Prim's and Kruskal's Algorithms, Single Source Shortest Paths **Space-Time Tradeoffs**: Introduction, sorting by Counting, Input Enhancement in String Matching (Horspool's Algorithm)	
UNIT – IV	**9 Hours**
Advanced Design and Analysis Techniques: Dynamic Programming- Elements of dynamic programming, longest common subsequence, Warshall - Floyd 's Algorithm, 0/1 Knapsack, Optimal Binary search trees **Amortized Analysis**: Aggregate analysis, The accounting method, The potential method, Dynamic tables.	
UNIT – V	**9 Hours**
Limitation of Algorithmic Power: Lower-Bound Arguments, P, NP and NP-Complete Problems **Backtracking:** n – Queens problem, Hamiltonian Circuit Problem, Subset – Sum Problem **Branch and Bound** - Assignment Problem, Travelling Salesman Problem	

LABORATORY

1. Implementation and execution of simple programs to understand running time analysis of non-recursive and Recursive algorithms
   a) Insertion Sort
   b) Tower of Hanoi problem.
2. Implement Merge sort algorithm using divide and conquer. Determine the time required to sort the elements. Repeat the experiment for different values of n and plot a graph of the time taken versus input size.
3. Implement Topological Ordering using Vertex deletion method
4. Develop programs to implement single source shortest paths using greedy technique
5. Application of Dynamic Programming strategy to solve Optimization Problems- Implementation of Warshall – Floyd's algorithms
6. Develop a program to implement string pattern matching algorithms using
   a) Brute force string pattern matching and
   b) Horspool's, algorithm using input enhancement technique and
   represent the comparative study of both algorithms in graph based on input and time taken.
7. Application of Backtracking strategy to solve combinatorial problems: Implementation of N- Queen's problem

Course Outcomes:

After going through this course, the students will be able to:

CO1	:	Apply the basic concepts of algorithm design and analysis. Use mathematical techniques to evaluate and compare the efficiency of algorithms
CO2	:	Design efficient algorithms using appropriate algorithmic strategies. Select the right technique based on problem characteristics
CO3	:	Analyze the efficiency of algorithms and justify the performance using time complexity. Compare alternative solutions using asymptotic analysis
CO4	:	Evaluate algorithmic approaches to solve real-world optimization and decision problems effectively

Reference Books

1. Anany Levitin, "Introduction to the Design and Analysis of Algorithms", Person Education, 3rd Edition, 2021, ISBN-13: 9780137541133
2. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, "Introduction to Algorithms", The MIT Press, Cambridge, Massachusetts London, England, 4th Edition, 2022, ISBN: 9780262046305
3. George T. Heineman, —Learning Algorithms: A Programmer's Guide to Writing Better Code, O'Reilly Media Inc 1st Edition, 2021, ISBN: 9781492091066
4. Lekh Raj Vermani and Shalini Vermani, —An Elementary Approach to Design and Analysis of Algorithms, World Scientific Publishing Europe Ltd., 2019, ISBN-13:978-1786346759

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
4.	**LAB:** Conduction of laboratory exercises, lab report, observation, and analysis (30 Marks), PBL progress review (10 marks) and lab test (10 Marks) adding up to 50 Marks. **THE FINAL MARKS WILL BE 50 MARKS**	**50**
	MAXIMUM MARKS FOR THE CIE	**150**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**
RUBRIC FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	**CONTENTS**	**MARKS**
Evaluation will be conducted through problem-/project-based learning activities specific to the program or laboratory. The assessment rubrics will be mapped to the respective course outcomes and shared with students along with the problem statements.		**50**

SEMESTER: II						
AGILE SOFTWARE DEVELOPMENT **Category: Professional Core Course** **(Theory)**						
Course Code	:	**MCA223TB**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	:	**3:0:0**		**SEE**	**:**	**100 Marks**
Pedagogical Hours(L:T:P:SL)	:	**45:0:0:45**		**CIE duration**	**:**	**2.00 Hours**
Total credits	:	**3**		**SEE Duration**	**:**	**3.00 Hours**

UNIT – I	**9 Hours**
Agile Software Development: Agile methodology frameworks for software Development Scrum Framework: What is Agile software Development? What is Scrum? Why should you choose Scrum? Foundation of Scrum, Pillars of Empiricism, Scrum Values, identifying a Scrum Team; Introducing the Scrum Master; Working with the Product Owner; Getting acquainted with the Developers **Case Study:** Demonstration of converting Software Requirements into Product Backlog using AI tools	
UNIT – II	**9 Hours**
Scrum Events- Getting ready to Sprint; Deciding on the Sprint duration; Setting the Sprint Goal Defining Done as working and potentially shippable software; Starting the Sprint with Sprint Planning- for the Scrum Master, Product Owner and Developers. Keeping on the right track with the Daily Scrum, Inspecting the product during a Sprint Review, Inspecting the team with the Sprint Retrospective **Case Study:** AI-assisted Sprint Planning	
UNIT – III	**9 Hours**
Scrum artifacts- Understanding the value of the Scrum Artifacts, Creating, Managing and refining product backlog; Making a commitment- the Product Goal- Creating backlog items as user stories, Creating Product backlog items as features, Refining the Product Backlog, Creating and Managing the Sprint Backlog; Producing a Product Increment. **Case Study:** Creating User Stories and Acceptance Criteria using AI Tools. Demonstration of Product Backlog prioritization using **MoSCoW** and **INVEST** principles.	
UNIT – IV	**9 Hours**
Scrum in Action - Planning and Estimating with Scrum: Choosing an Estimation Scale, Creating a Baseline. Playing Planning poker, Estimating backlog items using a Bucket method, Envisioning the product journey with a product roadmap; Sprint Journey: Refining the Product Backlog, learning how to apply order to the product backlog. Scrum master duties for Product Backlog Refinement. Tracking progress with a Scrum Board, Managing defects, Canceling the Sprint. **Case Study:** AI-assisted Estimation and Product Roadmap generation, Demonstration of Scrum Board implementation and use of Agile Metrics.	
UNIT – V	**9 Hours**
Facets of Scrum: Software development practices for Scrum- Source control model for continuous integration, Continuous delivery and continuous deployment, leveraging testing methods for Scrum, Applying Scrum to remote teams, Managing technical debt, Scaling Scrum. **Forward Deployed Engineering:** Introduction, FDE Operating Loop, Three Non-negotiable Characteristics, The FDE Mindset.	

Case Study: AI-assisted Code Review, Implementation of CI/CD pipeline using **GitHub Actions**		
Course Outcomes: After going through this course, the student will be able to:		
CO1	:	Apply Agile practices in managing real-world projects.
CO2	:	Analyze project scenarios to select the most appropriate Agile framework for optimal delivery of business value
CO3	:	Assess performance using Agile metrics to ensure project alignment with objectives.
CO4	:	Design a comprehensive Agile workflow tailored to a specific project, integrating user stories, iterative plans, and prioritization strategies.

Reference Books

1. Fred Heath, The Professional Scrum Master (PSM I) Guide, Packt Publishing 2021, ISBN 13: 978-1800205567
2. Susheela Hooda Vandana Mohindru Sood Yashwant Singh, Sandeep Dalal and Manu Sood, Agile Software Development, Trends, Challenges and Applications Wiley Publications, 2023, ISBN:9781119896395
3. Srinivas Bommena, Shipping Enterprise AI - The Forward Deployed Engineer's Playbook, Kindle edition, May 2026, ASIN : B0H2D6RV8L.
4. Clinton Keith, Agile Game Development with Scrum, Addison-Wesley Professional,2020, ISBN:978-0-321-61852-8

Web References

https://www.scrum.org/;
https://docs.github.com/;https://platform.openai.com/docs

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**

SEMESTER: II						
CYBER SECURITY **Category: Professional Elective Course (Group C)** **(Theory)**						
Course Code	:	**MCA225C1**		**CIE**	:	**100 Marks**
Credits: L:T:P	:	**3:0:0**		**SEE**	:	**100 Marks**
Pedagogical Hours(L:T:P:SL)	:	**45:0:0:45**		**CIE duration**	:	**2.00 Hours**
Total credits	:	**3**		**SEE Duration**	:	**3.00 Hours**

UNIT – I	**9 Hours**
Introduction: Defining Cyberspace and Cyber security, Standard of Good Practice for Information Security, NIST Cyber security Framework. **System Access:** System Access Concepts, User Authentication, Password-Based Authentication, Possession-Based Authentication, Biometric Authentication, Risk Assessment for User Authentication, Access Control, Customer Access	
UNIT – II	**9 Hours**
Phishing: Introduction, Phishing – Methods of Phishing, Phishing Techniques, Spear Phishing, Types of Phishing scams, Phishing Toolkits and Spy Phishing, Phishing Countermeasures. **Identity Theft**-Personally Identifiable Information (PII), Types of Identity Theft, Techniques of ID theft, Countermeasures, how to efface your online identity	
UNIT – III	**9 Hours**
Tools and Methods used in Cybercrime: Introduction, Proxy Server and Anonymizers, Steganography, DOS and DDOS attack, SQL injection, Buffer Overflow, Attacks on wireless and mobile networks, mobile devices **Testing and Tools: Social Engineering Attacks:** How a social engineering attack works, conducting a social engineering attack, common attacks used in penetration testing, preparing yourself for face-to-face attacks, defending against social engineering attacks	
UNIT – IV	**9 Hours**
Blockchain: History, types of blockchain, Consensus, Decentralization using blockchain, methods of decentralization, roots of decentralization, blockchain and full ecosystem decentralization, Smart contracts, decentralized organizations, Plot forms for decentralization	
UNIT – V	**9 Hours**
Analysis of Malicious Executables and Detection Techniques: Introduction, Malicious Code Classification System, Malware Behavior Analysis, Conventional Detection Systems, Classifying Executables by Payload Function, Result and Discussion **Detection and Analysis of Botnet Attacks Using Machine Learning Techniques**: Introduction, Botnet Architecture, Methodology Adopted. **Understanding Computer Forensics and Forensics of Handheld Devices** Introduction, Phases of Digital Forensics, Demystifying Artificial Intelligence in the Digital World, Application of Machine Learning in Digital Forensics Investigations, Implementation of Artificial Intelligence in Forensics, Pattern Recognition Using Artificial Intelligence, Applications of AI in Criminal Investigations	

Course Outcomes: After going through this course the student will be able to:		
CO1	:	Apply cybersecurity principles to secure digital systems.
CO2	:	Analyze cyber threats and identify suitable solutions to address security issues in cyber systems.
CO3	:	Demonstrate the use of cybersecurity techniques to enhance system security.
CO4	:	Design secure solutions for real-world cybersecurity challenges

Reference Books
1. William Stallings, Effective Cybersecurity: A Guide to Using Best Practices and Standards, Addison-Wesley Professional, 2018, ISBN-13: 978-0134772806
2. Nina Godbole Sunit Belapure, Cyber Security, Wiley India, 2012, ISBN: 9788126521791
3. Imran Bashir, "Mastering Blockchain", Packet Publishing Ltd. 3rd Edition 2020 ISBN:978-1-78883-904-4
4. Shilpa Mahajan, Mehak Khurana Vania Vieira Estrela, "Applying Artificial Intelligence in Cybersecurity Analytics and Cyber Threat Detection", Wiley, ISBN: 9781394196449

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**

SEMESTER: II						
Gen-AI **Category: Professional Elective Course (Group C)** **(Theory)**						
Course Code	:	**MCA225C2**		**CIE**	:	**100 Marks**
Credits: L:T:P	:	**3:0:0**		**SEE**	:	**100 Marks**
Pedagogical Hours(L:T:P:SL)	:	**45:0:0:45**		**CIE duration**	:	**2.00 Hours**
Total credits	:	**3**		**SEE Duration**	:	**3.00 Hours**

UNIT – I	**9 Hours**
Basics of Generative AI: Introduction to AI, NLP foundation, Types of Generative Models (Autoregressive, Variational, Adversarial) Applications of Generative AI (Art and Creativity, Image and Video Generation, Text Generation, Music Composition), Generative AI vs Discriminator AI, Popular Generative Models - GPT, DALL·E, StyleGAN	
UNIT – II	**9 Hours**
Introduction to Prompt Engineering: Core Prompting Techniques, Prompt Structure & Patterns Prompt Robustness & Safety, Retrieval & Contextual Grounding, Multi-Agent & Modular Patterns. Prompt Control & Conditioning, Prompt Optimization & Maintenance, The Future of Prompt Engineering	
UNIT – III	**9 Hours**
Overview of Large Language Models: LLMs and their architecture. Benefits, Challenges and Limitations. Best LLM and frameworks - GPT-4, Gemini, Lang Chain; Evolution of generation– GANs, GAN architecture - Generator and Discriminator networks, Types of GANs - DCGAN, WGAN, StyleGAN, Cycle GAN and attention mechanisms, Variable Auto Encoders	
UNIT – IV	**9 Hours**
Advanced NLP Techniques: Understanding BERT architecture and pre-training objectives. Fine- tuning BERT for downstream NLP tasks. Exploration of advanced Transformer architecture and techniques. **Applications and Future Directions:** Real-world applications of large language models. Challenges and limitations of current approaches. Emerging trends and future directions in Generative AI..	
UNIT – V	**9 Hours**
Security aspects in Gen AI: Misuse of Generated content, Adversarial Attacks, IP Theft, Bias Detection and Mitigation, Data Privacy and Security, Fairness, Transparency, and accountability, Regularity and legal considerations in LLMs, Ethical Principles in AI.	

Course Outcomes:
After going through this course, the student will be able to:

CO1	:	Apply the concepts and principles of Gen-AI for diverse AI applications
CO2	:	Demonstrate proficiency in using Gen- AI tools, features, frameworks and models for real-world Scenarios
CO3	:	Generate text, audio, image, and video content using LLM Models and Frameworks
CO4	:	Examine the ethical, security, and legal challenges associated with generative AI tools and models

Reference Books
1. Dr. Priyanka Singh, Hariom Singh, Generative AI Essentials: Unlocking creativity and innovation with generative AI, First Edition, Copyright © BPB Publications, India ISBN: 978-93-65897-074, 2025
2. Numa Dhamani and Maggie Engler, Introduction to Generative AI, First Edition, Manning Shelter Island, ISBN: 9781633437197, 2024
3. David Foster, Generative Deep Learning, O'Reilly Media, 2023, ISBN-978-1098134181
4. Ben Auffarth, Generative AI with Lang Chain: Build Large Language Model (LLM) apps with Python, ChatGPT and other LLMs, Packt Publishing, 2023, ISBN-978-1835083468
5. Amit Jain, Prompt Engineering: From Basics to Brilliance - Master Prompt Engineering through 46 Core Concepts | Beginner-Friendly | Step-by-Step Learning | Simple & Practical Guide, Jain Publication, ISBN-10 9356193479**,** ISBN-13 978-9356193475

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**

SEMESTER: II						
EXTENDED REALITY **Category: Professional Elective Course (Group C)** **(Theory)**						
Course Code	:	**MCA225C3**		**CIE**	:	**100 Marks**
Credits: L:T:P	:	**3:0:0**		**SEE**	:	**100 Marks**
Pedagogical Hours(L:T:P:SL)	:	**45:0:0:45**		**CIE duration**	:	**2.00 Hours**
Total credits	:	**3**		**SEE Duration**	:	**3.00 Hours**

UNIT – I	**9 Hours**
Introduction to Virtual and Augmented Reality: What is VR/AR about? – The perfect Virtual Reality, The Simulation of the World, Suspension of Disbelief, Motivation. Virtual Reality- Technology-centered characterizations of VR, VR as an Innovative Kind of Human-Computer Interaction, Mental Aspects of the VR Experience. VR Systems, Augmented Reality- Introduction, Registration, Visual Output- Handheld Devices, AR Systems, Mixed Reality (MR) and the XR Spectrum. Presence, Immersion and Interaction, Introduction to OpenXR and Cross-platform XR Standards, Role of Artificial Intelligence in XR: AI-assisted XR development and applications	
UNIT – II	**9 Hours**
Virtual Worlds: Requirements on 3D Object Representations for Virtual Worlds, Creation of 3D Models, including AI-assisted 3D asset generation, Preparation of 3D Models for VR/AR, Integration of 3D Models into VR/AR Runtime Environments. Surface models- Polygonal Representations, Polygons, Polygon Meshes, Physically Based Rendering (PBR) Materials, Solid Models- Level of Detail (LOD), Polygon Budget and Asset Optimization, 3D Asset Formats and Asset Pipeline, Appearance, Materials, Textures and Shaders	
UNIT – III	**9 Hours**
Unity 3D Game engine: Working in Unity- Getting Around in Unity, Working with Game Objects, Working with Components, Working with Prefabs, Working with Scenes, Managing Assets, Building Unity Projects, Accessing Preferences, Installing Unity Packages, Physics materials, Mesh colliders, Box collider, Materials, Textures, XR Plugin Management, OpenXR Support in Unity. XR Interaction Toolkit (Overview), Introduction to Unreal Engine, Actor–Component Model and Blueprint Visual Scripting (Concepts only)	
UNIT – IV	**9 Hours**
Animation and Object Behavior: Keyframe Animation, Physics based Animation of Rigid Bodies, Object Behavior, Behavior and Animation in Scene Graphs. XR Interaction Techniques: Controller-based Interaction, Hand Tracking (Introduction), Object Interaction (Grab, Throw and Use), Locomotion Techniques (Teleportation, Continuous Movement and Snap Turn), World-space User Interface. Lighting, Sound, Background, Special purpose systems- Virtual Humans, AI-enabled Virtual Characters and Conversational AI (Introduction), Particle Systems, Terrain, Vegetation	
UNIT – V	**9 Hours**
VR/AR Input Devices and Tracking: Fundamentals of XR Input Devices and 6DoF Tracking, Camera based Tracking- Marker-based Methods, Marker-free Tracking, Inside-out Tracking and Introduction to SLAM. Augmented Reality Tracking: Plane Detection and Image Tracking. XR Input Devices: Introduction to Hand Tracking and Eye Tracking. Performance Optimization: Frame Rate Budget, Stereo Rendering and Building & Deploying XR Applications.	

Course Outcomes: After going through this course, the student will be able to:		
CO1	:	Apply the concepts and principles of Extended Reality for developing immersive applications.
CO2	:	Analyze Extended Reality technologies and development methodologies to address application requirements
CO3	:	Implement Extended Reality solutions using appropriate tools, techniques and frameworks
CO4	:	Develop immersive Extended Reality applications by integrating XR technologies, development platforms and optimization techniques

Reference Books

1. Ralf Doerner, Wolfgang Broll, Paul Grimm, Bernhard Jung, "Virtual and Augmented Reality (VR/AR)- Foundations and Methods of Extended Realities (XR)", © Springer, ISBN 978-3-030-79061-5 ISBN 978-3-030-79062-2 (eBook)

2. Paris Buttfield-Addison, Jon Manning, Tim Nugent, "Unity Development Cookbook-Real Time Solutions from Game Development to AI", O'Reilly Media, Inc., August 2023, ISBN: 9781098113711

3. Suman Dutta, "Immersive Realm of Extended Reality", bpb publications 2024 Edition, ISBN: 9789355517227 eISBN: 9789355519450

4. Web links and Video Lectures (e-Resources)
   - Unity Learn – Unity Essentials
   - Unity Learn – VR Development Pathway
   - Unity Learn – Mobile AR Development Pathway
   - Unity XR Interaction Toolkit Documentation
   - Unity AR Foundation Documentation (ARCore/ARKit)
   - Khronos OpenXR Documentation

   Epic Developer Community – Unreal Engine Blueprint Visual Scripting and VR Development Tutorials

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**

SEMESTER: II						
SOFTWARE TESTING **Category: Professional Elective Course (Group C)** **(Theory)**						
Course Code	:	**MCA225C4**		**CIE**	:	**100 Marks**
Credits: L:T:P	:	**3:0:0**		**SEE**	:	**100 Marks**
Pedagogical Hours(L:T:P:SL)	:	**45:0:0:45**		**CIE duration**	:	**2.00 Hours**
Total credits	:	**3**		**SEE Duration**	:	**3.00 Hours**

UNIT – I	**9 Hours**
Basics of Software Testing: Basic Definitions, Evolution of Software Testing, Goals of Software Testing, Levels of Testing, Model for Software Testing, Software testing life cycle, Software testing methodologies: Waterfall testing, Agile Testing. Software failure case studies	
UNIT – II	**9 Hours**
Testing techniques **Black-Box Testing Techniques:** Boundary Value Analysis, Equivalence Class Testing, State Table-Based Testing, Decision Table-Based Testing, Cause-Effect Graphing Based Testing, Error Guessing, **White-Box Testing Techniques**: Need, Basis Path Testing, Graph Matrices, Loop Testing, Data Flow Testing, Mutation Testing, Static Testing, Progressive vs. Regressive Testing, Regression Testing Techniques.	
UNIT – III	**9 Hours**
Model-Driven Test Design (MDTD): Software testing foundations, Testing activities, Coverage Criteria, Test Design, Test Automation, Test Execution, Test Evaluation, Software Testing Guidelines, Importance of MDTD **Test Automation:** Need for Automation, Categorization of Testing Tools, Selection of Testing Tools, Guidelines for Automated Testing, Test Automation Framework: JUnit, Data-Driven Tests	
UNIT – IV	**9 Hours**
Managing the Testing Process: Test Organization, Structure of Testing Group, Test Planning, Detailed Test Design and Test Specifications, Testing Metrics for Monitoring and Controlling the Testing Process, **Cross-Platform Software Testing** Cross-Platform Software Testing: Understanding Cross-Platform Testing, Need, Challenges, Step- By-Step to Cross-Platform Testing, Cross-Platform Testing vs Cross-Device Testing, Overview of some best Cross-Platform Testing Tools in the Market(Appium, Eggplant)	
UNIT – V	**9 Hours**
Multiplatform Software Testing: Objectives, Concerns, Background on Testing in a Multiplatform Environment, Workbench, Input, Do Procedures and Tasks, Check Procedures, Output, Guidelines, Cross-Platform Software Testing Vs. Multiplatform Software Testing **Testing AI-Based Systems Overview**: Specification of AI-Based Systems, Test Levels for AI Based System, Testing data for testing AI-based Systems, What makes systems Non-Deterministic	

Course Outcomes: After going through this course the student will be able to:		
CO1	:	Apply various software testing techniques to ensure software quality in real-world scenarios.
CO2	:	Analyze test automation frameworks and tools to select the most suitable approach for diverse testing needs.
CO3	:	Evaluate software testing processes, metrics, and strategies to ensure software quality.
CO4	:	Design software testing solutions using appropriate testing methods, automation, and AI-based testing practices.

Reference Books
1. Naresh Chauhan, SOFTWARE TESTING Principles and Practices, OXFORD UNIVERSITY PRESS,2016, ISBN-13 : 978-0199465873
2. Paul Ammann, Jeff Offutt, Introduction to Software Testing, Cambridge University Press,2nd Edition.2016, ISBN-13:978-1107172012
3. Paul C. Jorgensen, Byron DeVries, "Software Testing, A Craftsman's Approach", Auerbach Publications, 5th Edition, 2021, ISBN-13:978-0367358495.
4. Iosif Itkin , Iuliia Emelianova, Dmitrii Degtiarenko, Anna-Maria Lukina, "Introduction to AI Testing: Guide to ISTQB® CT-AI Certification, Kindle Edition, BCS, The Chartered Institute for IT, September 2025
5. Jason Arbon, Testing AI: Engineering Confidence in Non-Deterministic Systems, Kindle Edition, ASIN : : B0H3MLR5BN, July 2026
WebLink : https://nptel.ac.in/courses/106105150

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**

SEMESTER: II						
RESEARCH METHODOLOGY **Category: NPTEL Online Course** **(Theory)**						
Course Code	:	**MIM426T**		**CIE**	:	
Credits: L:T:P	:	**2:0:0**		**SEE**	:	
Pedagogical Hours(L:T:P:SL)	:	**:0:0:**		**CIE duration**	:	
Total credits	:	**2**		**SEE Duration**		
This course is indicative only and it is subject to change based on the courses running at that time by NPTEL						
Duration of the ONLINE Course - 8 Weeks						
Week 1: A group discussion on what is research; Overview of research **Week 2**: Literature survey, Experimental skills **Week 3**: Data analysis, Modelling skills **Week 4**: Technical writing; Technical Presentations; Creativity in Research **Week 5**: Creativity in Research; Group discussion on Ethics in Research **Week 6**: Design of Experiments **Week 7**: Intellectual Property **Week 8**: Department specific research discussions						
Reference Books:						
1. Krishnaswami, K.N., Sivakumar, A. I. and Mathirajan, M., Management Research Methodology, Integration of Principles, Methods and Techniques, 17th Impression, Pearson India Education Services Pvt. Ltd, 2018. ISBN: 978-81-7758-563-6 2. William M. K. Trochim, James P. Donnelly, The Research Methods Knowledge Base, 3rd Edition, Atomic Dog Publishing, 2006, ISBN: 978-1592602919 3. Kothari C.R., Research Methodology Methods and Techniques, 4th Edition, New Age International Publishers, 2019, ISBN: 978-93-86649-22-5. 4. Levin, R.I. and Rubin, D.S., Statistics for Management, 8th Edition, Pearson Education: New Delhi, 2017, ISBN-13- 978-8184957495.						
GENERAL GUIDELINES						
1. NPTEL is an acronym for National Programme on Technology Enhanced Learning which is an initiative by seven Indian Institutes of Technology (IIT Bombay, Delhi, Guwahati, Kanpur, Kharagpur, Madras and Roorkee) and Indian Institute of Science (IISc) for creating course contents in engineering and science. 2. NPTEL is offering online certification courses through its portal - https://swayam.gov.in/nc_details/NPTEL 3. Enrollment to courses and exam registration can be done in ONLINE mode only. The link is available on NPTEL website http://nptel.ac.in/ 4. Students need to enroll for the NPTEL course and clear the exam. 5. In case students fail to get the certificate, they need to enroll for the same course once again, in the subsequent NPTEL semester and clear the exam. 6. If the same course is not offered by NPTEL (i.e. if the same course is not re-run) in the subsequent semester by NPTEL, the students need to write letter seeking permission from the Counsellor, HoD and Dean Academics with further approval from BoS Committee to take alternative course from the list announced by NPTEL. 7. Exam is conducted by NPTEL.						

SEMESTER: II						
INNOVATIVE DESIGN THINKING LAB **Category: Design Thinking/Skill Lab** **(Practice)**						
Course Code	**:**	**MCA427DI**		**CIE**	:	**50 Marks**
Credits: L:T:P	:	**0:0:1**		**SEE**	:	**50 Marks**
Pedagogical Hours(L:T:P:SL)	:	**0:0:30:0**		**CIE duration**	:	**2.00 Hours**
Total credits	:	**1**		**SEE Duration**	:	**2.00 Hours**

Phase – I

Empathy: The Empathy phases of the process are focused on understanding the experiences, emotions and motivations of others. Designers use specific empathy methods to learn more about the needs of the users for whom they are designing.
Methods: Interviewing Probes and Observations

Phase – II

Define: The Define phase of the process is focused on developing a point of view about the need of your user. During this stage of process, designers narrow from lots of information to a statement that is inspiring and specific.
Methods: Empathy Mapping, Point of View.

Phase – III

Ideate: The Ideate phase of the process is focused on generating as many solutions to a problem as possible. Once many solutions have been generated, students will select one to move forward to prototyping.
Methods: Brainstorming and Selection

Phase – IV

Prototype: The Prototype phase is where designers construct representation of their solutions. These representations are intended to elicit feedback and answer specific questions about a concept.
Methods: Improve, Rapid and Experiential Prototyping

Phase – V

Test: The Test phase of the process is focused on getting specific feedback about how ideas can improve. It is important to remember during this phase that prototypes are imperfect, but feedback is gift.
Methods: Testing

Course Outcomes:
After going through this course the student will be able to:

CO1	:	Analyze problems using different modes of thinking to understand their underlying causes.
CO2	:	Acquire adductive reasoning to find new problems
CO3	:	Develop innovative solutions to problems by applying creative thinking techniques
CO4	:	Apply human centric approaches while developing new solutions, products or services.

Reference Books

1. https://onlinecourses.nptel.ac.in/noc22_mg32/preview

Guidelines for Design Thinking Lab:	
1.	The Design Thinking Lab (DTL) is to be carried out by a team of two-three students.
2.	Each student in a team must contribute equally in the tasks mentioned below
3.	Each group has to select a theme that will provide solutions to the challenges of societal concern. The topics should be in line with the Sustainable Development Goals (SDG)
4.	The above five stages specified will be evaluated in three phases
5.	For every Phase of evaluation, the committee constituted by the department along with the coordinators would evaluate for CIE. The committee shall consist of respective coordinator & two senior faculty members as examiners. The evaluation will be done for each student separately.
6.	The team should prepare a Digital Poster and a report should be submitted after incorporation of any modifications suggested by the evaluation committee.

Scheme of Continuous Internal Examination (CIE-Lab)

The evaluation of the work will be done by the committee appointed by the director, Dept of MCA. The student should submit a report on the Case Study solved under the theme Evaluation will be carried out in THREE Phases.

Phase	**Activity**	**Marks**
I	Review I	10
II	Review II	20
III	Review III	15
	Final Report	05
	MAXIMUM MARKS FOR CIE	**50**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Lab)

The evaluation will be done by Internal and External examiners through Exhibition Mode.

The following weightage would be given for the exhibition

Q.NO.	**CONTENTS**	**MARKS**
1	Presentation through posters	15
2	Demonstration of the Prototype	25
3	Viva-voce	10
	MAXIMUM MARKS FOR SEE	**50**

SEMESTER: III						
FULL STACK APPLICATION DEVELOPMENT **Category: Professional core** **(Theory + Practice)**						
Course Code	:	**MCA261IA**		**CIE**	:	**100 + 50 Marks**
Credits: L:T:P	:	**3:0:1**		**SEE**	:	**100 + 50 Marks**
Pedagogical Hours(L:T:P:SL)	:	**45:0:30:45**		**CIE duration**	:	**2.00 + 2.00 Hours**
Total credits	:	**4**		**SEE Duration**	:	**3.00+3.00 Hours**

UNIT - I	**9 Hours**
Introduction to Full Stack Development: Understanding Full Stack Development, Technologies associated with Full Stack **Introduction to the MERN stack:** Introduction, The MVC architectural Pattern, MERN Components React, Node.js, Express, MongoDB, Advantages of MERN, Isomorphic; **Node.js** - event-driven programming, JavaScript closures Node modules, Common JS modules, Node.js core modules, Node.js third-party modules, Node.js file modules Developing Node.js web application, connecting and mounting middleware;	
UNIT - II	**9 Hours**
Introduction to MongoDB: Introduction to NoSQL, Introducing MongoDB, MongoDB sharding, **MongoDB CRUD operations**-Creating a new document, Creating a document using insert(), Creating a document using update(), Creating a document using save(); **Introduction to Mongoose**-Introducing Mongoose ,Connecting to MongoDB, Understanding Mongoose schemas, Creating the user schema and model, Registering the User model, Creating new users using save(), Finding multiple user documents using find(), Reading a single user document using find One(), Updating an existing user document Deleting an existing user document; **Extending your Mongoose schema**-Defining default values, Using schema modifiers, Predefined modifiers, Custom setter modifiers, Custom getter modifiers	
UNIT - III	**9 Hours**
Understanding React and Web Server: Server setup, NVM Node JS, Project, NPM, Express, Build time JSX compilation- Separate Script File, Transform, Automate, React Library, React Components-React classes, Composing components, passing data- using properties, property validation, using children's Dynamic composition; **Understanding React State:** React State: Setting state, Event handling, communicating from child to parent, Stateless components, Designing Components-state vs props, component hierarchy communication, Stateless components	
UNIT - IV	**9 Hours**
Building Express Web Application - Introduction to Express, Installing Express, Creating your first Express application, The application, request and response objects-The application object, The request object, The response object, External middleware **Building RESTful APIs and Mongo DB:** REST , HTTP method as Actions, Request Object, Response objects, Middleware, **GraphQL**: Field Specification, Graph Based, Single Endpoint, Strongly Typed, Introspection, Libraries, The About API, GraphQL Schema File,	
UNIT - V	**9 Hours**
The list API , The create API, Using the LIST API, Using the Create API, Error Handling **Working with React Router and forms:** Routing Techniques, Simple Routing, Route parameters, Route Query String, Programmatic Routing, Nested Routes, Browser history, React Forms, Filter form, Get API, Edit page, UI Components, update API, Delete API.	

LABORATORY

1. Demonstrate Node .Js Application to perform CRUD operation for online Book Cart
2. Write a node.js program using Express framework to accept user name, Branch, Semester, from web page and display the information as below
   a) Handle both get and post methods
   b) Branch should be underlined
   c) Name should be in bold face.
3. Design a resume of a job aspirant using React components like Classes and Functions. Style the resume by applying CSS
4. Build student registration portal using Entities like component, state and props
5. Design and implement a React Form that collects user input for name, email, and password. Validate the form using Regular Expression.
6. Deploy connectivity between React and Node Application for Inventory Management system
7. Develop a MongoDB query for comparison selectors, Logical Selectors for Company database
8. Execute aggregation pipeline and its operation to illustrate text search on catalog data collection
9. Design an employee Management system using RESTFULL APIs in React
10. Create a React application using react-router-dom with multiple pages (Home, About, Contact).

Course Outcomes:

After going through this course the student will be able to:

CO1	:	Demonstrate the core concepts of the Model-View-Controller (MVC) architecture and its role in structuring web applications
CO2	:	Apply the MVC pattern in building applications using the MERN stack components
CO3	:	Develop full-stack web applications using the MERN stack
CO4	:	Analyze the structure and functionality of web applications by implementing the MVC framework using the MERN stack

Reference Books

1. Subramanian, V. (2019). Pro MERN Stack: Full Stack Web App Development with Mongo, Express, React, and Node (2nd ed.). Apress. ISBN: 9781484243904
2. Hoque, S. (2022). Full-Stack React Projects: Learn MERN Stack Development by Building Modern Web Apps Using MongoDB, Express, React, and Node.js (2nd ed.). Packt Publishing. ISBN: 9781801070636
3. Ackermann, P. (2023). Full Stack Web Development: The Comprehensive Guide (Rheinwerk Computing). Rheinwerk Computing. ISBN: 9783969108830
4. Osmani, A. (2023). Learning JavaScript Design Patterns: A JavaScript and React Developer's Guide (2nd ed.). O'Reilly Media. ISBN: 9781098139872

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS. Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
4.	**LAB:** Conduction of laboratory exercises, lab report, observation, and analysis (30 Marks), PBL progress review (10 marks) and lab test (10 Marks) adding up to 50 Marks. **THE FINAL MARKS WILL BE 50 MARKS**	**50**
	MAXIMUM MARKS FOR THE CIE	**150**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	SEE THEORY TOTAL	**100**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Lab)		
Q.NO.	**CONTENTS**	**MARKS**
Evaluation will be conducted through problem-/project-based learning activities specific to the program or laboratory. The assessment rubrics will be mapped to the respective course outcomes and shared with students along with the problem statements		**50**

SEMESTER: III						
DEVOPS AUTOMATION **Category: Professional core** **(Theory + Practice)**						
Course Code	:	**MCA262IA**		**CIE**	:	**100 + 50 Marks**
Credits: L:T:P	:	**3:0:1**		**SEE**	:	**100 + 50 Marks**
Pedagogical Hours(L:T:P:SL)	:	**45:0:30:45**		**CIE duration**	:	**2.00 + 2.00 Hours**
Total credits	:	**4**		**SEE Duration**	:	**3.00+3.00 Hours**

UNIT - I	09 Hours
The DevOps Culture: Getting started with DevOps **Docker Fundamentals:** Discovering Docker, What and why of Docker, Building a Docker Application. Understanding Docker - Docker's Architecture, The Docker Daemon, The Docker Client, Docker Registries, The Docker Hub	
UNIT - II	**09 Hours**
Docker and Development: Using Docker as a lightweight Virtual Machine - From VM to Container, Saving and restoring your work, Environments as processes, Building Images, Running Containers. Volumes—a persistent problem: Docker volumes: Problems of persistence, Day-to-day Docker: Staying ship-shape - Running Docker without sudo, Detaching containers without stopping them, Using Portenier to manage your Docker daemon, Are you in a Docker container?	
UNIT - III	**09 Hours**
Docker and DevOps: Continuous Integration - Docker Hub automated builds, More efficient builds, Containerizing your CI process - Running the Jenkins master within a Docker container, Containing a complex development environment, Upgrading your containerized Jenkins server safely. Continuous delivery - Interacting with other teams in the CD pipeline, Facilitating deployment of Docker images, configuring your images for environments, Upgrading running containers	
UNIT - IV	**09 Hours**
First steps with Docker and Kubernetes: Creating, running, and sharing a container image, Setting up a Kubernetes cluster - Running a Local Single-Node Kubernetes Cluster with Minikube, Setting up an alias and command-line completion for kubectl, Running the first app on Kubernetes - Deploying your Node.js app, Accessing your web application, The logical parts of your system, Examining what nodes your app is running on, Introducing the Kubernetes dashboard	
UNIT - V	**09 Hours**
Pods: Introducing Pods, Creating pods from YAML or JSON descriptors, organizing pods with labels, Listing subsets of pods through label selectors, Annotating pods, Using namespaces to group resources, Stopping and removing pods	

LABORATORY
1. Build a Docker Container from a Custom Dockerfile.
2. Develop a Multi-Stage Dockerfile for Optimized Application Deployment.
3. Containerize and Deploy an AI-Enabled Python Flask Application.
4. Implement Version-Controlled Application Deployment Using Git and Podman
5. Demonstrate AI-Assisted CI/CD Integration Using Jenkins in Docker.
6. Deploy an Automated Container Build and Security Scanning Pipeline.
7. Deploy and Scale a Containerized Web Application on Kubernetes Using Minikube.
8. Create and Manage Kubernetes Workloads Using YAML Manifests.
9. Monitor Kubernetes Clusters Using Dashboard and CLI Tools.

Course Outcomes: After going through this course the student will be able to:		
CO1	:	Demonstrate the use of Docker for container creation and management.
CO2	:	Implement containerized development environments using Docker to ensure consistent workflows.
CO3	:	Develop automated CI/CD pipelines using Jenkins and Docker Hub to streamline integration and deployment processes.
CO4	:	Deploy containerized applications on Kubernetes and manage pods, labels, and namespaces for efficient workload orchestration

Reference Books

1. Gaurav Agarwal, "Modern DevOps Practices: Implement, secure, and manage applications on the public cloud by leveraging cutting-edge tools", Packt Publishing, 2nd Edition, 2024, ISBN–9781805121824

2. Ian Miell, Aidan Hobson Sayers, "Docker in Practice", Manning Publications, 2nd Edition, 2019, ISBN–9781617294808

3. Marko Lukša, "Kubernetes in Action", Manning Publications, 2nd Edition, 2018, ISBN–9781617293726

4. Brendan Burns, Joe Beda, and Kelsey Hightower, "Kubernetes: Up and Running", 2nd Edition, 2019, ISBN–978-1-492-04653-0

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)

CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**

Sl.No.	COMPONENTS	MARKS
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
	LAB: Conduction of laboratory exercises, lab report, observation, and analysis (30 Marks), PBL progress review (10 marks) and lab test (10 Marks) adding up to 50 Marks. **THE FINAL MARKS WILL BE 50 MARKS**	**50**
	MAXIMUM MARKS FOR THE CIE	**150**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	SEE THEORY TOTAL	**100**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Lab)		
Q.NO.	**CONTENTS**	**MARKS**
Evaluation will be conducted through problem-/project-based learning activities specific to the program or laboratory. The assessment rubrics will be mapped to the respective course outcomes and shared with students along with the problem statements		**50**

SEMESTER: III						
AI PRODUCT & PROJECT MANAGEMENT **Category: Professional Core Course** **(Theory)**						
Course Code	**:**	**MCA233TB**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Pedagogical Hours(L:T:P:SL)	**:**	**45:0:0:45**		**CIE duration**	**:**	**2.00 Hours**
Total credits	**:**	**3**		**SEE Duration**	**:**	**3.00 Hours**

UNIT - I	**9 Hours**
Understanding the Infrastructure and Tools for Building AI Products: Understanding the difference AI, ML, DL, Learning Types in ML, Order of Optimal Process Flow, Deployment Strategies, **Model Development and Maintenance for AI Products:** Understanding the stages of NPD, when is a model ready for market, Deployment, Testing and troubleshooting, Refreshing models	
UNIT - II	**9 Hours**
Building an AI-Native Product: Stages of AI product development, **Productizing the ML Service:** Understanding the differences between AI and traditional software products, Consistency and AIOps/MLOps–reliance and trust, Performance evaluation – testing, retraining and hyperparameter tuning, Feedback loop **Benchmarking Performance, Growth Hacking, and Cost:** Value metrics – a guide to north star metrics, KPIs and OKRs, Hacking – product-led growth, The tech stack – early signals, managing costs and pricing – AI is expensive	
UNIT - III	**9 Hours**
Product-Market Validation for AI-First SaaS: SaaS and Its Evolution, AI as a Service (AIaaS), Product-Market-Technology (P-M-T) and Validation Framework, Five-Step AI-First SaaS Validation Framework **AI Product Strategy:** Product Discovery, Product Requirements Analysis, Product	
UNIT - IV	**9 Hours**
Go-to-Market Strategy for an AI Startup: GTM foundations; Go-to-Market Strategy for AI (as a) Solution, AI-as-a-Service, and AI-as-a-Toolkit; Determining the Pricing Model and Packaging That Appeals to Developers, Developing a Marketing Strategy	
UNIT - V	**9 Hours**
Project Management Principles: Project Management Mindset, 12 Project Management Principles:Stewardship, team, stakeholders, value, systems thinking, leadership, tailoring, quality, complexity, risk, adaptability, sustainability. **Life Cycles & Development Approaches:** Predictive, adaptive (Agile), hybrid; Considerations for a Development Approach Selection, Delivery Cadence, Project Management Focus Areas: Initiating, Planning, Executing, Monitoring & Controlling,	

Course Outcomes: After going through this course the student will be able to:		
CO1	:	Apply project management principles and suitable development approaches to AI product projects
CO2	:	Analyze AI products based on performance, cost, customer needs, and market requirements
CO3	:	Develop strategies for validating, launching, marketing, and managing AI-based products.
CO4	:	Examine processes involved in developing AI-based products.

Reference Books

1. Irene Bratsis, The AI Product Manager's Handbook, Packt Publisher, 1st Edition, February 2023, ISBN 9781804612934.
2. Adhiguna Mahendra, AI Startup Strategy: A Blueprint to Building Successful Artificial Intelligence Products from Inception to Exit, Apress, 2023, ISBN-13 (pbk): 978-1-4842-9501-4 ISBN-13 (electronic): 978-1-4842-9502-1, https://doi.org/10.1007/978-1-4842-9502-1
3. PMI, PMBOK® Guide, ANSI/PMI 99-001-2025 8th Edition., 2025, ISBN: 978-1-62825-829-5 (paperback), ISBN: 978-1-62825-830-1 (ebook)
4. Justin Norman, Peter Skomoroch, Mike Loukides, Product Management for AI, O'Reilly Media, Inc, February 2021, ISBN: 9781098104191.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**

SEMESTER: III						
BUSINESS INTELLIGENCE & VISUALIZATION PRACTICES **Category: Professional elective (Group E)** **(Theory + Practice)**						
Course Code	**:**	**MCA234EA**		**CIE**	**:**	**100 + 50 Marks**
Credits: L:T:P	**:**	**3:0:1**		**SEE**	**:**	**100 + 50 Marks**
Pedagogical Hours(L:T:P:SL)	**:**	**45:0:30:45**		**CIE duration**	**:**	**2.00 + 2.00 Hours**
Total credits	**:**	**4**		**SEE Duration**	**:**	**3.00+3.00 Hours**

UNIT - I	**9 Hours**
Introduction to Business Intelligence and Data Visualization **Business Intelligence and its Architecture:** BI by other names, How BI provides business value, BI Market, Battle Scars, The Research, Best Practices for successful Intelligence **Components of BI Architecture**: Operational and Source Systems, Data Transfer – from Operational to Data Warehouse, Data Warehouse, DW tables, Technology platform **Defining Data Visualization**: The Components of Understanding – The Importance of Conviction - Visualization work flow – The importance of process – Process in practice – Different tools for Data Visualization	
UNIT - II	**9 Hours**
Data Types and Power BI **Working with data**: Data Literacy – Data Assets and Tabulation types – Data types – Statistical Literacy – Data Acquisition – Data Examination **Overview of Power BI**: Understanding Power BI – Features – Connect to Different Data Sources Using Power BI and Data Modeling	
UNIT - III	**9 Hours**
Data Handling and Interaction **Handling data**: Data Analysis and Expressions (DAX) – Calculated Columns – Representation using Data Columns **Interactivity – Features of Interactivity:** Data Adjustments and Presentation Adjustments – Influencing Factors and Considerations	
UNIT - IV	**9 Hours**
Usage of Data for visualization **Data Representation**: Introducing visual encoding – Chart Types – Influencing Factors and Considerations – Visualization using Graphs, Plots, Charts and Geospatial Maps using Power BI	
UNIT - V	**9 Hours**
Visualization and Query handling: Visualization Literacy **–** Viewing: Learning to See – Creating: The Capabilities of the Visualizer **Creating Reports and Publishing Reports** – Design Dashboards and Publishing using Gateways, Basic power BI queries	

LABORATORY

1. Analysis of revenue in sales dataset and build dashboard with following criteria:
   a. Create a choropleth map (fill the map) to spot the special trends to show the state which has the highest revenue.
   b. Create a line chart to show the revenue based on the month of the year.
   c. Create a bin of size 10 for the age measure to create a new dimension to show the revenue.
   d. Create a donut chart view to show the percentage of revenue per region by creating zero access in the calculated field.
   e. Create a butterfly chart by reversing the bar chart to compare female, male revenue based on product category.
   f. Create a calculated field to show the average revenue per state, display profitable and non-profitable state.
2. Analysis of Amazon Prime Dataset and create the dash board with following criteria:
   a. Create a Donut chart to show the percentage of movie and tv shows
   b. Create a area chart to shows by release year and type.
   c. Create a horizontal bar chart to show Top 10 genre
   d. Create a map to display total shows by country
   e. Create a text sheet to show the description of any movie/movies.
3. Analyze the Cancer Patient Dataset and create the following
   a. What is the distribution of cancer stages across different age groups and genders?
   b. Which treatment type is most common among patients with late-stage cancer?
   c. What is the survival rate trend across different types of cancer (if survival/outcome data is available)?
4. Based on the Employee Data set, develop a Dashboard for the following:
   a. What is the employee attrition rate across departments and job roles over time?
   b. Which departments have the highest average salary and what is their employee satisfaction level (if available)?
   c. How does employee experience (years of service) affect promotion and resignation trends?
5. Using the Unemployment Analysis Data set, build the dashboard
   a. Which states have consistently high or low unemployment rates across different months?
   b. What is the trend of unemployment rate by region and educational qualification (if available)?
   c. How does the unemployment rate change seasonally and what are the peak unemployment periods?
6. Build a Dashboard for the Mobile Sales Dataset
   a. Which brand or model has the highest total revenue and units sold across all regions?
   b. What is the sales trend across different quarters/months and which products peak during which periods?
   c. What is the profit margin comparison between different mobile models or brands?
7. Analyze and Develop a dashboard with the following questions
   a. How does academic stress correlate with sleep duration and mental health status among students?

b. Which gender or age group reports higher levels of anxiety, depression, or stress?
c. What is the distribution of mental health conditions across different types of academic programs or courses?

8. Develop a dashboard to analyze waste generation and recycling patterns in urban areas of India. The dashboard will focus on key metrics such as waste types, recycling rates, disposal methods, and community engagement in sustainability practices
9. Develop a dashboard to monitor wildlife populations and their habitats in India, focusing on key metrics such as population trends, habitat conditions, and human impact factors (like deforestation, pollution, or poaching).
10. Develop a dashboard and service to monitor and optimize urban mobility by analyzing traffic patterns, public transportation usage, and air quality in Indian cities.

Course Outcomes:

After going through this course, the student will be able to:

CO1	:	Apply the principles of Business Intelligence and Data Visualization techniques on the different data types.
CO2	:	Demonstrate the relevant visual encoding techniques like chart, graph, plot etc for real time examples using visualization tools
CO3	:	Perform data analysis, graphical representation, and interpretation to various phenomena in real life applications
CO4	:	Create interactive reports and dashboards using various visualization techniques

Reference Books

1. Cindi Howson, Successful Business Intelligence, McGraw-Hill Publications, 2nd Edition,2017, E-ISSN: 0-07-149851-6, ISBN:978-9339213220
2. Andy Kirk, Data Visualization, A Handbook for Data Driven Design, SAGE Publications, India Pvt Ltd, first edition, 2016, ISBN 978-1-4739-1213-7, ISBN 978-1-4739-1214-4 (pbk)
3. Suren Machiraju, Suraj Gaurav, Power BI Data Analysis and Visualization, De|GPRESS, 2018, ISBN 978-1-5474-1678-3 , e-ISBN (PDF) 978-1-5474-0072-0 , e-ISBN (EPUB) 978-1-5474-0074-
4. Jonathon Schwabish, Better data visualizations: a guide for scholars, researchers, and wonks, Columbia University Press, Feb 2021, LCCN 2020017814 (print) | LCCN 2020017815 (ebook) | ISBN 9780231193108 (hardback) | ISBN 9780231193115 (trade paperback) | ISBN 9780231550154 (eBook)

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
4.	**LAB:** Conduction of laboratory exercises, lab report, observation, and analysis (30 Marks), PBL progress review (10 marks) and lab test (10 Marks) adding up to 50 Marks. **THE FINAL MARKS WILL BE 50 MARKS**	**50**
	MAXIMUM MARKS FOR THE CIE	**150**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**
RUBRIC FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	**CONTENTS**	**MARKS**
Evaluation will be conducted through problem-/project-based learning activities specific to the program or laboratory. The assessment rubrics will be mapped to the respective course outcomes and shared with students along with the problem statements.		**50**

SEMESTER: III						
INTERNET OF THINGS **Category: Professional elective (Group E)** **(Theory + Practice)**						
Course Code	:	**MCA234EB**		**CIE**	:	**100 + 50 Marks**
Credits: L:T:P	:	**3:0:1**		**SEE**	:	**100 + 50 Marks**
Pedagogical Hours(L:T:P:SL)	:	**45:0:30:45**		**CIE duration**	:	**2.00 + 2.00 Hours**
Total credits	:	**4**		**SEE Duration**	:	**3.00+3.00 Hours**

UNIT - I	**9 Hours**
Introduction to Internet of Things: Fundamentals of Electronics and devices for Internet of Things: Rectification process, Diode characteristics, Digital electronics, Transistor behavior and Oscillators Physical and Logical design of IoT Technologies that enable Internet of Things Applications and Use cases, IoT Deployment Levels. Network and Communication, Standards related to Internet of Things, Protocols in Internet of things	
UNIT - II	**9 Hours**
Programming with Arduino UNO and ESP32 : Understanding the Eco system of Arduino, Pin- out configuration, Digital input and output, Analog input and output, working with sensors and actuators. Arduino serial communication. Communication interfaces (SPI and I2C) wired and wireless communication with Arduino using Bluetooth modules	
UNIT - III	**9 Hours**
Programming with Raspberry Pi and Pico W: Understanding the eco system of Raspberry Pi4 and Pico W, Pinout configuration, python modules like Rpi.GPIO and gpiozero. Digital input and output, working with sensors and actuators. Raspberry Pi or Pico serial communication. Communication interfaces (SPI and I2C).wired and wireless communication with raspberry Pi or Pico W. Serial communication from raspberry Pi4 to Arduino and vice versa. Monitoring and Controlling between raspberry pi.	
UNIT - IV	**9 Hours**
Working with IoT Platform and Dashboard: Configuring any hosted service like blynk or Adafruit or Things board with development boards, sensors and actuators. **Development of Local IoT Dashboards:** Integrating node-red with sensors and actuators, customizing node red using java script and integrating alerting services with node red. Setup and configure MQTT publisher , broker and subscriber using Raspberry pi 4 and exp32 generate visualization and analytics based on the data logged about the sensors, Working with things board and setting up IoT dashboards locally and configure sensor, actuators and development boards with things board	
UNIT - V	**9 Hours**
Case study: Industrial IoT Applications, Study on Industrial IoT application related to controlled environment agriculture, energy monitoring and harvesting, production system, vision based applications, water quality monitoring, health monitoring of plants, machines or humans and location based applications	

LABORATORY

Practice Lab :

Fundamentals of Electronics using SEELab3 kit and Introduction to variety of devices and development boards used to develop IoT Applications

Full wave rectifier using PN junction: Refer Section 3.3 in the SEElab3 kit manual Diode V-I functional analysis Refer Section 3.13 in the SEElab3 kit manual Logic gates: Refer Section

3.11 in the SEElab3 kit manual PNP & NPN transistor nature: Refer Section 3.13 and 3.15 in the SEElab3 kit manual IC555 oscillator: Refer Section 3.6 in the SEElab3 kit manual Identifying the IoT Kit elements: sensors, actuators and development boards and other accessories, Study about the principle of operations, operating conditions, cost, tolerance and durability of different devices in the kit

1. Develop the weather station using Arduino UNO and Esp32 integrating different sensors Note: Students should develop their own models and integrate sensors to mimic the weather station. Custom wind speed and direction calculation, Visualization about the same with history of data to be provided
2. Demonstrate MQTT publisher, MQTT BROKER and MQTT Subscriber using Raspberrypi4 ESP32 or Pico W and Esp32. Save the sensor data published in a SD card and also integrate visualization and basic analytics.
   Note: The data to be published could be related to water quality parameters or data could be related to plant, machine or person health parameters
3. Develop a custom IoT dashboard or things board using node-red and integrate alerting services based on the sensor values
   Note : node red installation, customization of widget and integrating the alerting services has to be demonstrated by integrating to those application other than scenarios given in program1 and program2

Course Outcomes:

After going through this course the student will be able to:

CO1	:	Apply fundamental programming skills to explore various libraries developed for different development boards like Arduino Uno, esp32, raspberry pi 4 or Raspberry Pico W to repeat monitor different sensor values, log the monitored data and control actuators.
CO2	:	Design automation scripts using different programming and scripting languages like bash script/python script/JavaScript to monitor and control the sensor and actuators to integration with development boards
CO3	:	Develop custom IoT dashboards and integrate sensors, actuators and alerting services using tools like node red and Things board
CO4	:	Analyze various IioT integration protocols, standards and tools available as hosted service for IoT data visualization and analysis using hosted cloud based IoT platform and dashboard services

Reference Books
1. Arshdeep Bahga, Vijay Madisetti, Internet of Things: A Hands-on Approach, Orient Blackswan Private Ltd, July 1st 2015, ISBN : 8173719543.
2. Hands-on ESP32 with Arduino IDE: Unleash the power of IoT with ESP32 and build exciting projects with this practical guide, Packt Publisher, Zulfiqar, 1st Edition, 2024.
3. Elector, The Official ESP32 Book, ISBN: 978-1-907920-63-9.
4. The Official Raspberry Pi Handbook, The Magpi Magazine, 2023.
URL resources https://nodered.org/ https://thingsboard.io/ https://www.raspberrypi.co https:// www.raspberrypi.com/documentation/microcontrollers/pico-series.html https://www.espressif.com/en/products/socs/esp32 https://csparkresearch.in/seelab3 https://www.arduino.cc/

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
4.	**LAB:** Conduction of laboratory exercises, lab report, observation, and analysis (30 Marks), PBL progress review (10 marks) and lab test (10 Marks) adding up to 50 Marks. **THE FINAL MARKS WILL BE 50 MARKS**	**50**
	MAXIMUM MARKS FOR THE CIE	**150**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**

RUBRIC FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	**CONTENTS**	**MARKS**
Evaluation will be conducted through problem-/project-based learning activities specific to the program or laboratory. The assessment rubrics will be mapped to the respective course outcomes and shared with students along with the problem statements.		**50**

SEMESTER: III						
CLOUD COMPUTING APPLICATIONS **Category: Professional elective (Group E)** **(Theory + Practice)**						
Course Code	**:**	**MCA234EC**		**CIE**	**:**	**100 + 50 Marks**
Credits: L:T:P	**:**	**3:0:1**		**SEE**	**:**	**100 + 50 Marks**
Pedagogical Hours(L:T:P:SL)	**:**	**45:0:30:45**		**CIE duration**	**:**	**2.00 + 2.00 Hours**
Total credits	**:**	**4**		**SEE Duration**	**:**	**3.00+3.00 Hours**

UNIT - I	**9 Hours**
Introduction & Concepts: Introduction to Cloud Computing: Characteristics of Cloud Computing, Cloud Models, Cloud Service Examples, Cloud-based Services & Applications. **Migrate into a Cloud:** Introduction, Broad Approaches to Migrating into the Cloud, The Seven-Step Model of Migration into a Cloud	
UNIT - II	**9 Hours**
Cloud Concepts & Technologies: Virtualization, Load Balancing, Scalability & Elasticity, Deployment, Replication, Monitoring, Software Defined Networking, Network Function Virtualization, Identity and Access Management, Service Level Agreements, Billing.	
UNIT - III	**9 Hours**
Cloud Application Design: Introduction, Design Considerations for Cloud Applications, Reference Architectures for Cloud Applications, Cloud Application Design Methodologies, Data Storage Approaches.	
UNIT - IV	**9 Hours**
Cloud Security: Introduction, CSA (Cloud Security Architecture) Authentication, Authorization, Identity & Access Management, Data Security, Key Management, Auditing. Virtual Machine Security, Security of Virtualization, Security risk posted by a management OS	
UNIT - V	**9 Hours**
Edge Computing Fundamentals- Finding the edge, edge computing characteristics, edge computing scenarios **Edge Computing Factors-** Edge computing networking, Edge devices, Fleet management, Perspectives and uniformity	

LABORATORY

1. Launch a Linux or Window Server by creating VPC, Route Table in a cloud
2. Create Storage space using Storage Services in cloud
3. Demonstrate the working of Load Balancer and Elastic Ips in cloud.
4. Demonstrate Identity and access management (IAM) in cloud
5. Create RDS Server and connect using MySQL Workbench
6. Deploy PHP Code on EC2 instance that retrieves data from RDS Server
7. Build static website and host application from desktop on cloud
8. Demonstrate VS code Integration with cloud
9. Deploy Java application by connecting to RDS Server in cloud
10. Demonstrate auto scaling group concept in cloud

Course Outcomes:

After going through this course, the students will be able to:

CO1	:	Apply cloud concepts and technologies to solve practical computing problems
CO2	:	Analyze cloud application design methodologies and approaches for building efficient cloud-based systems.
CO3	:	Design secure cloud services for cloud applications
CO4	:	Evaluate the effectiveness of cloud solutions, understanding its components and processes.

Reference Books

1. Arshdeep Bahga, Vijay Madisetti, Cloud Computing A Hands-on Approach", University Press, 2014, Edition, ISBN: 9788173719233
2. Thomas Erl, Eric Barc, Cloud Computing: Concepts, Technology, Security & Architecture, 2nd Edition – Pearson Paperback – 29 February 2024.ISBN 978-81-969-4321-9.
3. Rajkumar Buyya, Christian Vecciola, Thamarai Selvi, & Moreel Monroy, Mastering Cloud Computing, 2nd Edition, McGrawHILL Paperback – 1 June 2024.ISBN-13:978-93-5532- 950-9
4. Douglas Comer, The Cloud Computing Book: The Future of Computing Explained Paperback – Import, 24 July 2023, CRC PRESS ISBN-13: 978-0367706845
5. Haishi Bai, Boris Scholl, Edge Computing and Capability-oriented Architecture, CRC Press. Taylor and Francis Group, Edition 1, ISBN-13978-1032046723, 2024

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
4.	**LAB:** Conduction of laboratory exercises, lab report, observation, and analysis (30 Marks), PBL progress review (10 marks) and lab test (10 Marks) adding up to 50 Marks. **THE FINAL MARKS WILL BE 50 MARKS**	**50**
	MAXIMUM MARKS FOR THE CIE	**150**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**
RUBRIC FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	**CONTENTS**	**MARKS**
Evaluation will be conducted through problem-/project-based learning activities specific to the program or laboratory. The assessment rubrics will be mapped to the respective course outcomes and shared with students along with the problem statements.		**50**

SEMESTER: III						
AGENTIC AI **Category: Professional elective (Group E)** **(Theory + Practice) (Industry Elective)**						
Course Code	:	**MCA234ED**		**CIE**	:	**100 + 50 Marks**
Credits: L:T:P	:	**3:0:1**		**SEE**	:	**100 + 50 Marks**
Pedagogical Hours(L:T:P:SL)	:	**45:0:30:45**		**CIE duration**	:	**2.00 + 2.00 Hours**
Total credits	:	**4**		**SEE Duration**	:	**3.00+3.00 Hours**

UNIT - I	**9 Hours**
Introduction to Generative AI, GenAI in the Enterprise, The transformative potential of GenAI, **Overview of business applications**: Horizontal applications (cross-functional use cases), Vertical or domain-specific applications. **Introducing agentic AI systems:** The anatomy of agentic AI, Core components, Agent anatomy, Data stores and environment context, Key architectural features, The **GenAI Maturity Model:** The new agentic stack, Enabling agent communication from tools to collaboration, Agent internals (common to A and B for simplicity), The MCP server, The agent server, Challenges hindering production-grade GenAI	
UNIT - II	**9 Hours**
Essential Components of Intelligent Agents: Technical requirements, Knowledge representation in intelligent agents, Reasoning in intelligent agents, learning mechanisms for adaptive agents, Decision-making and planning in agentic systems, Enhancing agent capabilities with generative AI. Agent Lifecycle, Cross-Cutting Components in AI Agents, Types of Memory in Agentic AI, Conflicts, Memory Poisoning and Validation Techniques,	
UNIT - III	**9 Hours**
Agentic AI Architecture: Components and Interactions, **Defining an agent:** core concepts and capabilities, LLMs, Automated workflows, AI agents, The anatomy of an agent: Case study: Travel Planning Agent, Case study: Agentic Loan Processing System, Illustrative flow, loan application lifecycle, Multi-agent coordination via A2A , Data stores and environment context for agents. **Agent interaction models and key features:** Architectural features, Agent interaction models, Technical considerations for agentic architectures	
UNIT - IV	**9 Hours**
Multi-Agent Coordination Patterns: A strategic guide to implementing coordination patterns: Multi-agent systems: foundational coordination (Level 4), Advanced multi-agent coordination and self-correction (Levels 5–6), The Agent Router pattern (intent-based routing). **Task Delegation Frameworks:** Supervisor Architecture (centralized orchestration),Swarm Architecture, Multi-Agent planning.	
UNIT - V	**9 Hours**
Building Intelligent RAG Systems: Components of a RAG system, From embedding to search: Embedding, Vector stores, RAG Pipeline, Types of Chunking, Types of Retrieval, Corrective RAG, Agentic RAG. **Introduction to Agent Development Ecosystem:** Challenges of Building Agents from Scratch, Need for Orchestration, Evolution of agent frameworks (LangChain , LangGraph): How LangChain enables agent development, Exploring the LangChain architecture, Modular design and dependency management LangGraph, LangSmith, and companion tools, Exploring LangChain's building blocks, Running local models.	

LABORATORY

1. Design an AI assistant using an LLM with appropriate system and user prompts to perform a specified task.
2. Create an AI agent capable of invoking external tools (e.g., calculator, web search, or custom functions) to solve user queries.
3. Design and implement an agent that processes loan applications based on predefined eligibility rules and generates recommendations.
4. Extend a conversational agent to retain and utilize conversation history for context-aware interactions.
5. Build a document question-answering system using embeddings, a vector database, and an LLM.
6. Enhance the loan processing agent by integrating RAG to answer queries using loan policy documents.
7. Implement an agent router that classifies user requests and routes them to appropriate specialized agents.

Course Outcomes:

After going through this course the student will be able to:

CO1	:	Apply the concepts of Generative AI, Agentic AI, and agent architectures to understand enterprise AI applications.
CO2	:	Design workflows and multi-agent systems using Agentic AI principles for real-world applications.
CO3	:	Build AI agents using LLMs, workflows, tools, and RAG for solving real-world problems.
CO4	:	Asses Agentic AI solutions, architectures, and enterprise use cases to select suitable approaches for different application scenarios.

Reference Books

1. Arsanjani, J. P. Bustos, and T. Kuria, Agentic Architectural Patterns for Building Multi-Agent Systems: Proven design patterns and practices for GenAI, agents, RAG, LLMOps, and enterprise-scale AI systems, Packt Publishing, 1st Edition,2026. ISBN-13: 978-1806029570
2. A. Biswas, W. Talukdar, Building Agentic AI Systems: Create intelligent, autonomous AI agents that can reason, plan, and adapt. Birmingham, UK: Packt Publishing, 2025. ISBN: 978-1803238753
3. Ben Auffarth, Leonid Kuligin. Generative AI with LangChain: Build Production-Ready LLM Applications and Advanced Agents Using Python, LangChain, and LangGraph,2nd Edition, Packt Publishing, 2025. ISBN-13: 978-1-83702-201-4.
4. S. Raieli and G. Iuculano, Building AI Agents with LLMs, RAG, and Knowledge Graphs: A practical guide to autonomous and modern AI agents. Birmingham, UK: Packt Publishing, 2025. ISBN: 978-1835080382

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
4.	**LAB:** Conduction of laboratory exercises, lab report, observation, and analysis (30 Marks), PBL progress review (10 marks) and lab test (10 Marks) adding up to 50 Marks. **THE FINAL MARKS WILL BE 50 MARKS**	**50**
	MAXIMUM MARKS FOR THE CIE	**150**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**
RUBRIC FOR SEMESTER END EXAMINATION (LAB)		
Q.NO.	**CONTENTS**	**MARKS**
Evaluation will be conducted through problem-/project-based learning activities specific to the program or laboratory. The assessment rubrics will be mapped to the respective course outcomes and shared with students along with the problem statements.		**50**

SEMESTER: III						
DEEP LEARNING **Category: Professional elective (Group D)** **(Theory)**						
Course Code	**:**	**MCA263D2**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Pedagogical Hours(L:T:P:SL)	**:**	**45:0:0:45**		**CIE duration**	**:**	**2.00 Hours**
Total credits	**:**	**3**		**SEE Duration**	**:**	**3.00 Hours**

UNIT – I	**9 Hours**
Neural Networks: Introduction to NN, models of neuron and network architectures. Learning Processes: Different types of learning processes, Learning with and without teacher, Memory, statistical learning theory. **Single layer perceptron**: Adaptive filter problem, least mean square algorithm, learning rate, Learning rate, annealing techniques, perceptron and perceptron convergence theorem. **Multilayer Perceptron:** Back propagation algorithm, Sequential and batch modes of training, stopping criteria, XOR problem, and some numerical problems **Regularization Techniques:** Dropout and Batch Normalization	
UNIT - II	**9 Hours**
Convolutional Neural Networks: Introduction, Historical Perspective and Biological Inspiration. **Basic Structure of a Convolutional Network**: Padding, Strides, Typical Settings, The ReLU Layer, Pooling, Fully Connected Layers, The Interleaving Between Layers, Multiplications, Data Augmentation. **Training a Convolutional Network:** Back propagating Through Convolutions, Back propagation as Convolution with Inverted/Transposed Filter, Convolution/Back propagation as Matrix **Transfer Learning:** Introduction and Applications	
UNIT - III	**9 Hours**
Convolutional Architectures: AlexNet, ZFnet VGG, GoogLeNet, ResNet, EfficientNet (Introduction) Effect of Depth, Pretrained Models. **Applications of CNN:** Content based image retrieval, Object Localization, Object Detection, Natural Language and sequence learning, and Video classification	
UNIT - IV	**9 Hours**
Recurrent Neural Networks: Introduction and expressiveness of RNN. **Basic Structure of a RNN:** Language Modeling Example of RNN, Generating a Language Sample, Back propagation Through Time, Bidirectional Recurrent Networks, Multilayer Recurrent Networks. Long Short-Term Memory (LSTM), Gated Recurrent Units (GRUs) **Attention Mechanism:** Introduction. **Applications of Recurrent Neural Networks:** Automatic Image Captioning, Temporal Recommender Systems, Secondary Protein Structure Prediction, End-to-End Speech Recognition, Handwriting Recognition	

UNIT – V	**9 Hours**

Deep Reinforcement Learning: Introduction.

Stateless Algorithms: Multi-Armed Bandits, Naïve Algorithm, Greedy Algorithm, Upper Bounding Methods.

Basic Framework of Reinforcement Learning: Challenges of Reinforcement Learning, Role of Deep Learning.

Practical Applications of Deep Learning: Computer Vision, Natural Language Processing and Recommendation Systems.

Challenges in Deploying Deep Learning Models: Model Complexity, Interpretability and Scalability.

Course Outcomes:

After going through this course the student will be able to:

CO1	:	Apply deep learning concepts, neural network models and learning techniques to solve intelligent computing problems
CO2	:	Analyze deep learning architectures, learning algorithms and model performance for different application requirements
CO3	:	Assess deep learning solutions using appropriate neural network architectures, frameworks and optimization techniques
CO4	:	Develop intelligent applications by integrating deep learning models, learning techniques and deployment strategies to address real-world problems

Reference Books

5. Charu C. Aggarwal, “Neural Networks and Deep Learning: A Textbook”, 2nd Edition, Springer International Publishing, 2023, ISBN: 978-3-031-29642-0, eISBN: 978-3-031-29643-7.
6. Aston Zhang, Zachary C. Lipton, Mu Li, Alexander J. Smola, “Dive into Deep Learning”, Cambridge University Press, 2023, ISBN: 978-1009389433.
7. Anisha Udayakumar, Shriram K Vasudevan, Sini Raj Pulari “Advanced Deep Learning”, Wiley Publications. 2025, ISBN: 978-93-7060-695-5
8. Simon Haykin, “Neural Networks – A Comprehensive Foundation”, 2nd Edition, PHI, 2005, ISBN: 978-0131471399.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**

SEMESTER: III						
ADVANCED COMPUTER NETWORKS **Category: Professional elective (Group D)** **(Theory)**						
Course Code	**:**	**MCA263D3**		**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**		**SEE**	**:**	**100 Marks**
Pedagogical Hours(L:T:P:SL)	**:**	**45:0:0:45**		**CIE duration**	**:**	**2.00 Hours**
Total credits	**:**	**3**		**SEE Duration**	**:**	**3.00 Hours**

UNIT – I	**9 Hours**
Internet Protocol- Introduction, Error and Control Messages (ICMP): The Internet Control Message Protocol, Error Reporting vs Error Correction, Testing Destination Reachability and status, Echo Request and Reply Message Format. Classless and Subnet Address Extension (CIDR): Review of Relevant Facts, Proxy ARP, Subnet Addressing, Subnet Mask Representation, Broadcasting the Subnets, A Classless Addressing example	
UNIT - II	**9 Hours**
Wireless LANS and PANS – Fundamentals of WLAN's, 802.11 Standards, HIPERLAN Standard, Blue tooth specifications, Transport Protocol group, ZigBee Specification Wireless WANS and MANS – The Cellular Concept and Cellular Architecture- Capacity enhancement, Channel Allocation Algorithms	
UNIT - III	**9 Hours**
Mobile IP - Introduction, Mobility, Routing and Addressing, Mobile IP Characteristics, Overview of Mobile IP Operations, Mobile Addressing Details, Foreign Agent Discovery, Agent Registration, registration message format, communication with a foreign agent, datagram transmission and reception, two- crossing problem, communication with computers on the home network Private	
UNIT - IV	**9 Hours**
Advanced Internetwork Router Implementation: The Global Internet-Routing Areas, Inter domain Routing (BGP), IP Version 6(IPv6), Multiprotocol Label Switching (MPLS)- Destination Based forwarding, SCTP, Explicit Routing, Network Interconnection- NAT, VPN- Introduction.	
UNIT - V	**9 Hours**
SDN - Network Fundamentals and Technological Developments: Introduction traditional Network Architectures Limits of Traditional Network Architectures, SDN Network Overview, Advantages of an SDN Network, Main SDN Components, API North and South: The Pillars of SDN Orchestration, SDN Network Models, An Example of SDN Architecture in Action, Main SDN Controllers: An Overview, SDN Multi-Controller Network **Network Programming and Automation**: Introduction Network Programming and SDN Networks. Different Network Programmability Models, Imperative and Declarative Programming Models, Application Programming Interface **The OpenFlow Protocol:** Introduction, Benefits of OpenFlow protocol, OpenFlow protocol Versions, Components of an OpenFlow architecture, Management and Role of Tables in OpenFlow Switching, How OpenFlow works.	

Course Outcomes: After going through this course the student will be able to:		
CO1	:	Apply advanced networking concepts to establish and troubleshoot network communication.
CO2	:	Analyze protocols and addressing schemes to determine their applicability in different networking environments.
CO3	:	Select suitable networking protocols, architectures, for designing efficient network solutions.
CO4	:	Evaluate conventional and advanced network architectures, protocols, on performance.

Reference Books

1. Douglas E Comer, Internetworking with TCP/IP, Pearson Education India, 6th Edition, 2015, ISBN-13: 978-8180141744
2. C. Siva Ram Murthy, B. S. Manoj, Ad Hoc Wireless Networks Architecture and Protocols, Pearson Publication, 2018, ASIN : B0F6D37FB4
3. Ali Sadiqui and Moulay Rachid Filali, Fundamentals of Software-Defined (Towards Intelligent and Flexible Networks), CRC Press, January 30, 2026, ISBN 9781041154280.

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)

CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**

Sl.No.	COMPONENTS	MARKS
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)

Q.NO.	CONTENTS	MARKS
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**

SEMESTER: III						
PRINCIPLES OF UI/UX DESIGN **Category: Professional elective (Group D)** **(Theory)**						
Course Code	:	**MCA263D4**		**CIE**	:	**100 Marks**
Credits: L:T:P	:	**3:0:0**		**SEE**	:	**100 Marks**
Pedagogical Hours(L:T:P:SL)	:	**45:0:0:45**		**CIE duration**	:	**2.00 Hours**
Total credits	:	**3**		**SEE Duration**	:	**3.00 Hours**

UNIT - I	9 Hours
Elements of UX Design-I- Introduction, from product design to user experience design, designing for experience, User experience and the web, Building from bottom to top. **Strategy Plane:** Product Objectives, Business goals, Brand Identity, Success Metrics and User Needs, User Segmentation, Usability and User Research, Creating Personas **Scope Plane:** Defining the Scope, Functional specifications, Content requirements.	
UNIT - II	**9 Hours**
Elements of UX Design -II **Structure Plane:** Interaction Design, Conceptual Models, Error Handling, Information Architecture **Skeleton Plane**: Interface Design, Navigation Design, Information Design, Wireframes **Surface Plane:** Sensory Design, Making Sense of the Senses, Contrast and Uniformity, Internal and External Consistency, Color Palettes and Typography, Design Comps and Style Guides.	
UNIT - III	**9 Hours**
UI Design Process: Usability of Interactive Systems: Introduction, Usability Goals and Measures, Usability Motivation, Universal Usability, Guideline, principles, and theories Managing Design Processes: Introduction, Organizational Design to support Usability, The Four Pillars of Design, Development methodologies, Ethnographic Observation, Participatory Design, Scenario Development	
UNIT - IV	**9 Hours**
User Interface Evaluation and Interacting Styles Evaluating Interface Design: Introduction, Expert Reviews, Usability Testing and Laboratories, Survey Instruments, Acceptance tests, Evaluation during Active Use, Controlled Psychologically Oriented Experiments. Menu Selection, Form Filling and Dialog Boxes: Introduction, Task-Related Menu Organization, Single Menus, Combination of Multiple Menus, Content Organization, Fast Movement Through, Menus, Data Entry with Menus, Form Filling, Dialog Boxes and Alternatives, Audio Menus and Menus for Small Displays	
UNIT - V	**9 Hours**
Patterns For Effective Interaction Design **Using social media**: The Patterns-Editorial Mix, Personal Voices, Repost and Comment, Inverted Nano-pyramid, Sharing Widget, Content Leaderboard. **Going Mobile:** Patterns-Vertical Stack, Touch Tools, Bottom Navigation, Thumbnail-and-Text List, Loading Indicators, Richly Connected Apps. **Visual Style and Aesthetics**: Visual Design for Desktop applications, The Patterns Deep Background, Few Hues Many Values, Contrasting Font Weights Skins and Themes **Case Study:** To explore the UI/UX using AI/Digital tools(Uizard, UXpilot, UXMagic, Figma AI, FramerAI, etc), dark Patterns	

Course Outcomes: After going through this course the student will be able to:		
CO1	:	Apply the theoretical foundations of User Interface and User Experience design.
CO2	:	Design cross-platform UI and UX solutions using appropriate features, approaches, and patterns.
CO3	:	Apply UI and UX design skills to real-world applications
CO4	:	Evaluate UI/UX design processes and artefacts for building products

Reference Books

1. Jesse James Garrett, "The Elements of User Experience: User-Centered Design for the Web and Beyond", New Riders Publishers ,2 nd Edition, 2011, ISBN-13: 978-0321683687
2. Ben Shneiderman, Plaisant, Cohen, Jacobs, "Designing the User Interface", Pearson Education, 5th Edition, 2014, ISBN-10: 9332518734 ISBN-13: 978-9332518735
3. Bill Buxton, "Sketching User Experiences: Getting the Design Right and the Right Design",Morgan Kaufmann, 1st Edition,2007, ISBN-10: 0123740371 ISBN-13: 978-0123740373
4. Jenifer Tidwell, Charles Brewer, and Aynne Valencia, "Designing Interfaces- Patterns for Effective Interaction Design", O'Reilly®, 3 rd Edition, 2019, ISBN 978-1492051916
5. Jeff Gothelf and Josh Seiden, "Lean UX: Applying Lean Principles to Improve User Experience", O'Reilly Media , 3 rd Edition, 2021, ISBN 978-1492080359,

E-Resources

https://maze.co/collections/ux-ui-design/tools/#list
https://careerfoundry.com/en/blog/ux-design/dark-patterns-ux/

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component **[10 (Q) + 40 (T) + 50 (EL) = 100 marks]**		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**

SEMESTER: III

E-MARKETING
Category: Professional elective (Group D)
(Theory)

Course Code	**:**	**MCA263D5**	**CIE**	**:**	**100 Marks**
Credits: L:T:P	**:**	**3:0:0**	**SEE**	**:**	**100 Marks**
Pedagogical Hours (L:T:P:SL)	**:**	**45:0:0:45**	**CIE duration**	**:**	**2.00 Hours**
Total credits	**:**	**3**	**SEE Duration**	**:**	**3.00 Hours**

UNIT - I — **9 Hours**

Digital Marketing in Digital World: What Is Digital Marketing? , Evolution and Environment of Digital Marketing, Types of Digital Marketing, Digital Marketing Value,The Digital Marketing Mix, Digital Marketing Objectives, Digital Marketing Metrics and Analytics

Creating Value through Digital Marketing Strategy: Introduction to Digital Marketing Strategy, The Strategic Digital Marketing Planning Process, Basic Digital Marketing Strategies and Tactics, The Strategic Digital Marketing Plan

UNIT - II — **9 Hours**

Digital Consumer Behavior and Customer Relationship and Experience Management : Digital Consumer Behavior and Influences, Digital Customer Relationship Management, Digital Customer Experience (DCX) Management

Website Marketing Strategy: Website Evolution and Value, Website Marketing Objectives, Website Marketing Considerations, Strategies, and Tactics, Website Content Strategies, Website Marketing Analytics; User Experience (UX) and User Interface (UI) Considerations

Search Engine Optimization (SEO) Strategy: Evolution and Value of Search Engine Optimization, SEO Objectives, SEO Considerations, Strategies, and Tactics, SEO Analytics,

UNIT - III — **9 Hours**

Search Engine Marketing (SEM) Strategy : SEM Objectives, SEM Considerations, Strategies, and Tactics, SEM Content Strategies and Tactics,

Email Marketing Strategy: Evolution and Value of Email Marketing, Email Marketing Considerations, Strategies, and Tactics, Email Marketing Content and Design Strategies and Tactics, Email Marketing Analytics,

Mobile Marketing Strategy: Evolution and Value of Mobile Marketing, Mobile Marketing Considerations, Strategies, and Tactics, Mobile Marketing Content Strategies and Tactics, Mobile Marketing Analytics, AI-assisted Email Marketing and Campaign Automation (Introduction)

UNIT - IV — **9 Hours**

Digital Brand, Trust, and Reputation Management Strategy: Brand Switching Online, Digital Brand Positioning, Online Trust Building, Reputation Management.

Social media and Community Marketing Strategy: Evolution and Value of Social Media and Community Marketing, social media and Community Marketing Considerations, Strategies, and tactics, social media and Community Marketing Content Strategies and Tactics, Social Media and Community Marketing Analytics

Content Marketing Strategy: Evolution and Value of Content Marketing, Content Marketing Objectives, Strategies and Tactics, Content Planning and Distribution, Content Marketing Analytics.

UNIT – V	9 Hours
Digital Marketing Legal, Ethical, Privacy, and Security: Digital Marketing Legal and Ethical Framework, Digital Marketing Legal and Ethical Considerations, Strategies, and tactics **Trends and the Future of Digital Marketing,** Digital Marketing Trends and Future Considerations, Strategies, and Tactics	

Course Outcomes:

After going through this course the student will be able to:

CO1	:	Understand the core concepts, evolution and environment of digital marketing and strategies
CO2	:	Apply various digital marketing tools and techniques such as SEO, SEM, email, and mobile marketing to design strategic campaigns
CO3	:	Analyse digital consumer behaviour, website metrics and digital marketing analytics to evaluate campaign effectiveness
CO4	:	Design and formulate an integrated digital marketing strategy considering legal, ethical and branding factors.

Reference Books

1. Raj Sachdev, "Digital Marketing", Tata McGraw Hill , Copyright 2024, ISBN-13 : 978-1266214431
2. Seema Gupta, "Digital Marketing", Tata McGraw Hill, India, 3rd Edition, 2022, ISBN-9789355320483
3. Dave Chaffey, Fiona Ellis-Chadwick, "Digital Marketing- Strategy, Implementation and Practice", Pearson, 6th Edition, 2016, ISBN: 978-1-292-07761-1

RUBRIC FOR THE CONTINUOUS INTERNAL EVALUATION (CIE-Theory)		
CIE will consist of ONE Quiz (Q), TWO Tests (T), and ONE Experiential Learning (EL) component [10 (Q) + 40 (T) + 50 (EL) = 100 marks]		
Sl.No.	**COMPONENTS**	**MARKS**
1.	**QUIZ:** Quiz will be conducted in online/offline mode. **ONE QUIZ** will be conducted & will be evaluated for 10 marks, **THIS WILL BE CONSIDERED AS FINAL QUIZ MARKS.**	**10**
2.	**TESTS:** Students will be evaluated in test consisting of descriptive questions with different complexity levels (Revised Bloom's Taxonomy Levels: Remembering, Understanding, Applying, Analysing, Evaluating, and Creating). **THREE** tests will be conducted (Two regular tests and one optional improvement test). Each test will be evaluated for 50 Marks, adding up to 100 Marks. **FINAL TEST MARKS WILL BE REDUCED TO 40 MARKS.** **Students should score minimum 50% in TEST & QUIZ to clear CIE**	**40**
3.	**EXPERIENTIAL LEARNING:** Students will be evaluated for their creativity and practical implementation of the problem. **Phase I (25) & Phase II (25) ADDING UPTO 50 MARKS**.	**50**
	MAXIMUM MARKS FOR THE CIE	**100**

RUBRIC FOR SEMESTER END EXAMINATION (SEE-Theory)		
Q.NO.	**CONTENTS**	**MARKS**
1 & 2	Unit 1: Question 1 or 2	20
3 & 4	Unit 2: Question 3 or 4	20
5 & 6	Unit 3: Question 5 or 6	20
7 & 8	Unit 4: Question 7 or 8	20
9 & 10	Unit 5: Question 9 or 10	20
	MAXIMUM MARKS FOR THE SEE	**100**

SEMESTER: III						
Course Code	:	**MCA461P**	**MINOR PROJECT**	**CIE**	:	**100 Marks**
Credits L:T:P	:	**0:0:3**		**SEE**	:	**100 Marks**
Pedagogical Hours L:T:P:SL	:	**90**		**CIE Duration**	:	**2 Hours**
Total Credits	:	**3**		**SEE Duration**	:	**3 Hours**

Guidelines

1. Student can form a group of two to execute the Minor Project.
2. Students are required to select topics related to their PG Program Specialization, after extensive Literature Survey and analyzing the Research gaps.
3. Students will be assigned to guides in accordance with the expertise of the faculty.
4. Minor project topics could also be aligned to be implemented/executed based on any of the Centre of Excellence (CoE) / Center of Competence (CoC) domain. The details of these could be obtained by visiting the website https://rvce.edu.in/rvce-center-excellence
5. Minor project has to be implemented/executed in-house, using the resources available in the department/college/CoE/CoC.
6. Students have to note the periodic progress in the Minor Project Diary and report the work carried to their respective guides.
7. Students have to present the Minor project work to the departmental committee and only upon approval by the committee, the student can proceed to prepare and submit the hard copy of the final Minor project report.
8. The reports shall be printed on A4 size with 1.5 spacing and Times New Roman with font size 12, outer cover of the report (wrapper) has to be softbound in Ivory/White color for PG circuit Programs and Light Blue for Non-Circuit Programs.

Course Outcomes: After going through this course, the student will be able to

CO1	Analyze the research gaps, formulate the problem definition, conceptualize the objectives and design solution to cater to specific problems.
CO2	Apply higher order thinking skills and develop skill competencies specific to program specialization to implement real world problems with professional ethical standards.
CO3	Demonstrate the skill and knowledge by applying appropriate tools and techniques specific to their domain.
CO4	Communicate, work in teams and demonstrate the learning through oral presentations and report writing.

Scheme of Continuous Internal Evaluation (CIE):

The evaluation committee shall consist of Guide, Professor, Associate Professor/Assistant Professor. The committee shall assess and evaluate the presentation and the progress reports.

The evaluation criteria shall be as per the rubrics given below:

Reviews	Activity	Weightage
I	Approval of the selected topic, formulation of Problem Statement and Objectives along with Synopsis submission	10%
II	Demonstrate the skill and knowledge by applying appropriate tools/techniques to design solution specific to the problem.	30%
III	Demonstrates the work carried out through experimental results, analysis and testing; Exhibits writing and communication skills through presentations and report writing.	60%

Scheme for Semester End Evaluation (SEE):

The SEE examination shall be conducted by an external examiner (domain expert) and an internal examiner**. Evaluation shall be carried out in batches, not exceeding 6 students per batch.**

RUBRICS FOR SEMESTER END EXAMINATION

The SEE examination shall be conducted by an external examiner (domain expert) and an internal examiner.

Q.NO.	CONTENTS	MARKS
1	Write Up	**20%**
2	Demonstration of Minor Project Work	**60%**
3	Viva voce	**20%**

SEMESTER: IV						
Course Code	:	**MCA442P**	**MAJOR PROJECT**	**CIE**	:	**100 Marks**
Credits L-T-P	:	**0:0:10**		**SEE**	:	**100 Marks**
Pedagogical Hours L:T:P:SL	:	**0:0:300:0**		**CIE Duration**	:	**2 Hrs**
Total Credits	:	**10**		**SEE Duration**	:	**3 Hrs**

Guidelines

1. Major Project is to be carried out for a duration of 18 weeks
2. Students have to implement the Major Project individually.
3. Students are required to select topics related to their PG Program Specialization after extensive Literature Survey and analyzing the Research gaps.
4. Students will be assigned to guides in accordance with the expertise of the faculty.
5. Major project topics could also be chosen to be implemented/executed based on any of the 16 Centre of Excellence (CoE)/ 06 Center of Competence (CoC) domain. The details of these could be obtained by visiting the website https://rvce.edu.in/rvce-center-excellence
6. Major Project could be implemented in Industry/Research organizations after providing the letter of approval. Students can also implement Major Project, in-house using the resources available in the department/college/CoE/CoC.
7. Students have to adhere to the Project Presentation Schedule, record their periodic progress in the Major Project Diary and report the work carried to their respective guides.
8. It is mandatory for the students to present/publish their project work in National/International Conferences/Journals
9. Students have to present their Major Project work to the departmental committee and only upon approval by the committee, the student can proceed to prepare and submit the hard copy of the final Major Project report.
10. Major Project report has to be printed on A4 size with 1.5 spacing and Times New Roman with font size 12, outer cover of the report (wrapper) has to be soft bound in Ivory/White color for PG circuit Programs and Light Blue for Non-Circuit Programs.

Course Outcomes:

After going through this course, the student will be able to:

CO1	Analyze the research gaps, formulate the problem definition, conceptualize the objectives and design solution to cater to specific problems.
CO2	Apply higher order thinking skills and develop skill competencies specific to program specialization to implement solutions to real world problems, with professional ethical standards.
CO3	Demonstrate the skill and knowledge by applying appropriate tools and techniques specific to their domain.
CO4	Communicate, work in teams and demonstrate the learning through oral presentations and report writing.

Scheme of Continuous Internal Evaluation (CIE)		
The evaluation committee shall consist of Guide, Professor, Associate Professor/Assistant Professor. The committee shall assess and evaluate the presentation and progress reports.		
The evaluation criteria shall be as per the rubrics given below:		
Reviews	**Activity**	**Weightage**
I	Approval of the selected topic, formulation of Problem Statement and Objectives along with Synopsis submission.	10%
II	Demonstrate the skill and knowledge by applying appropriate tools/techniques to design solution specific to the problem.	30%
III	Demonstrates the work carried out through experimental results, analysis and testing; Exhibits writing and communication skills through presentations, report writing and paper publication.	60%

Scheme for Semester End Evaluation (SEE):		
Major Project SEE evaluation shall be conducted in two stages. This is initiated after fulfilment of submission of Project Report and CIE marks. Stage-1 Report Evaluation: Evaluation of Project Report shall be carried out by Guide and an External examiner. Stage-2 Project Viva-voce: Major Project Viva-voce examination is conducted after receipt of evaluation reports from Guide and External examiner.		
RUBRICS FOR SEMESTER END EXAMINATION		
SEE procedure is as follows:		
Report Evaluation	Internal Examiner: 100 Marks **(A)**	**Report Evaluation (A)+ (B) = 200/2 = 100 (C)**
	External Examiner: 100 Marks **(B)**	
Viva-Voce	Jointly evaluated by Internal Guide & External Examiners	**100 (D)**
	Total Marks = **(C+D)/2** = 200/2 =100	**100 Marks**

Note -.

i) 50% CIE is the prerequisite to appear for SEE

ii) Two hard bound dissertation reports are to be submitted.

iii) Certificate sheet having the signatures of Guide, Director and Principal must be included.

iv) Plagiarism report must be <=10% and to be included in the report.

SEMESTER: IV						
Course Code	:	**MCA443N**	**PROJECT INTERNSHIP**	**CIE**	:	**100 Marks**
Credits L-T-P	:	**0-0-6**	*(Practice)*	**SEE**	:	**100 Marks**
Pedagogical Hours L:T:P:SL	:	**0:0:180:0**		**CIE Duration**	:	**2 Hours**
Total Credits		**06**		**SEE Duration**	:	**3 Hours**

GUIDELINES

1. Students may opt to undergo internships in industries or research organizations such as BEL, DRDO, ISRO, NAL, etc.
2. Students must submit a letter from the industry/research organization clearly specifying the candidate's name and the duration of the internship. The letter must be printed on the organization's official letterhead and bear the signature of the authorized signatory.
3. The internship shall be undertaken on a full-time basis for a duration of six weeks after the completion of the II Semester final examinations and before the commencement of the III Semester.
4. Students may approach the Centres of Excellence (CoEs) in various domains and the Centres of Competence (CoCs) hosted by RVCE for registration and participation in relevant training/internship activities. Details may be obtained from the website: https://rvce.edu.in/rvce-center-excellence.
5. The internship must be related to the specialization area of the respective PG programme in which the student is enrolled.
6. Students undergoing internship training are advised to report their progress regularly and submit periodic progress reports/diaries to their respective guides.

Students shall present the internship activities carried out during the internship to the Departmental Committee. Upon approval by the Committee, the student may proceed with the preparation and submission of the final internship report. The report shall be printed on A4-size paper using Times New Roman font, size 12, with 1.5 line spacing, and shall be submitted in soft-bound form..

Course Outcomes:
After going through this course, the student will be able to

CO1	Explore the workplace, operating procedures of the department/company and its products, and other organizational concepts
CO2	Learn and improve writing and communication skills, research and technology, work in a team, and develop leadership skills
CO3	Apply higher order thinking skills - critical thinking, analysis, synthesis and evaluate complex problems to solve real world problems with professional ethical standards.
CO4	Develop and demonstrate skill competencies and knowledge specific to program specialization by applying appropriate tools and techniques.

Scheme of Continuous Internal Evaluation (CIE):
The evaluation committee shall consist of a Guide, Professor/Associate Professor and Assistant Professor. The committee shall assess the presentation and the progress reports in two reviews.

Reviews	Activity	Weightage
Review I	Ability to comprehend the functioning/operating procedures of the Organization/Departments. Application of Engineering knowledge, Critical thinking and analysis to solve problems.	40%
Review II	Demonstrates skill competencies, Resource Management and Sustainability. Exhibits writing and communication skills through presentations and report writing.	60%

Scheme for Semester End Evaluation (SEE): The SEE examination shall be conducted by an external examiner and an internal examiner. Evaluation shall be done in batches, not exceeding 6 students per batch.

Sl. No	Contents	Marks
1	Write Up	20%
2	Demonstration of Internship Work	60%
3	Viva Voce	20%

PROGRAMME OUTCOMES (PO)

MCA Graduates will be able to:

PO1 Foundation Knowledge: Apply knowledge of mathematics, programming logic and coding fundamentals for solution architecture and problem solving**.**

PO2 Problem Analysis: Identify, review, formulate and analyze problems for primarily focusing on customer requirements using critical thinking frameworks.

PO3 Development of Solutions: Design, develop and investigate problems with as an innovative approach for solutions incorporating ESG/SDG goals.

PO4 Modern Tool Usage: Select, adapt and apply modern computational tools such as development of algorithms with an understanding of the limitations including human biases.

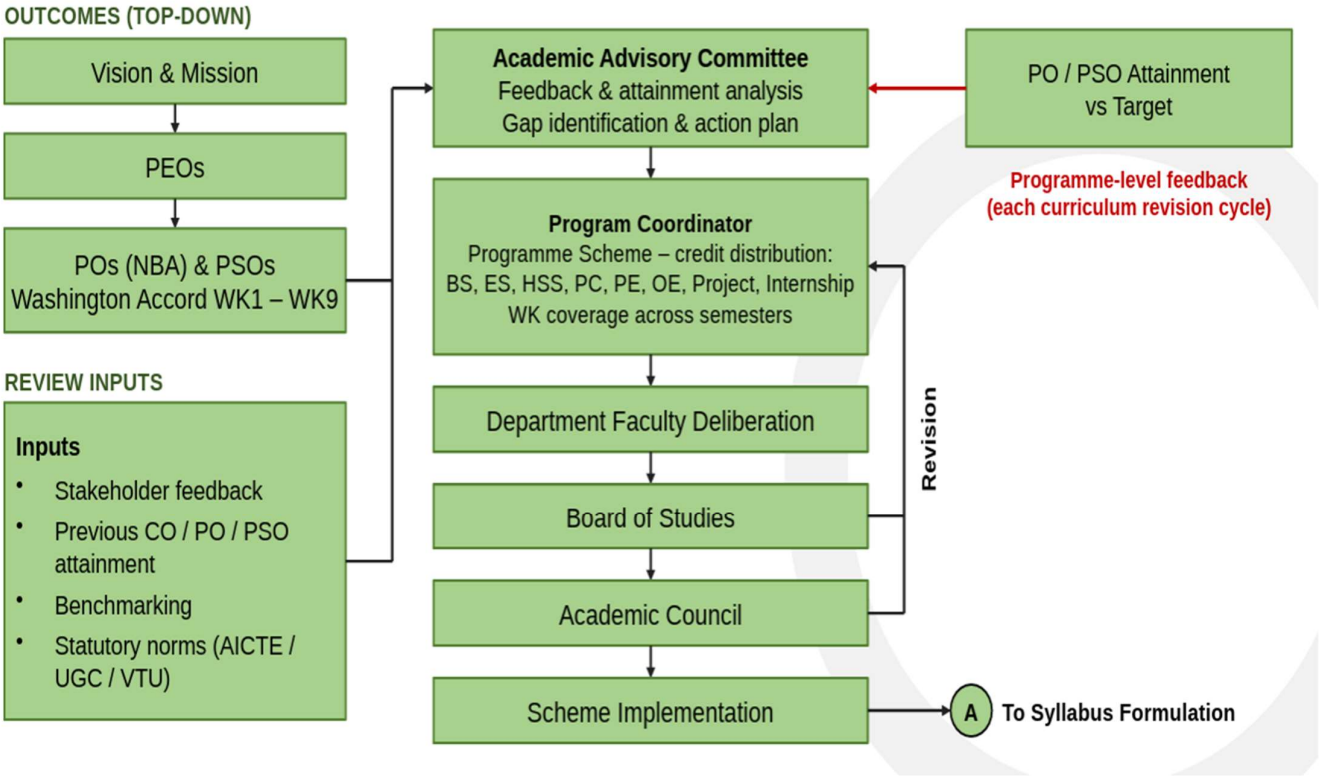
PO5 Individual and Teamwork: Function and communicate effectively as an individual or a team leader in diverse and multidisciplinary groups. Use methodologies such as agile.

PO6 Project Management and Finance: Use the principles of project management such as scheduling, work breakdown structure and be conversant with the principles of Finance for profitable project management

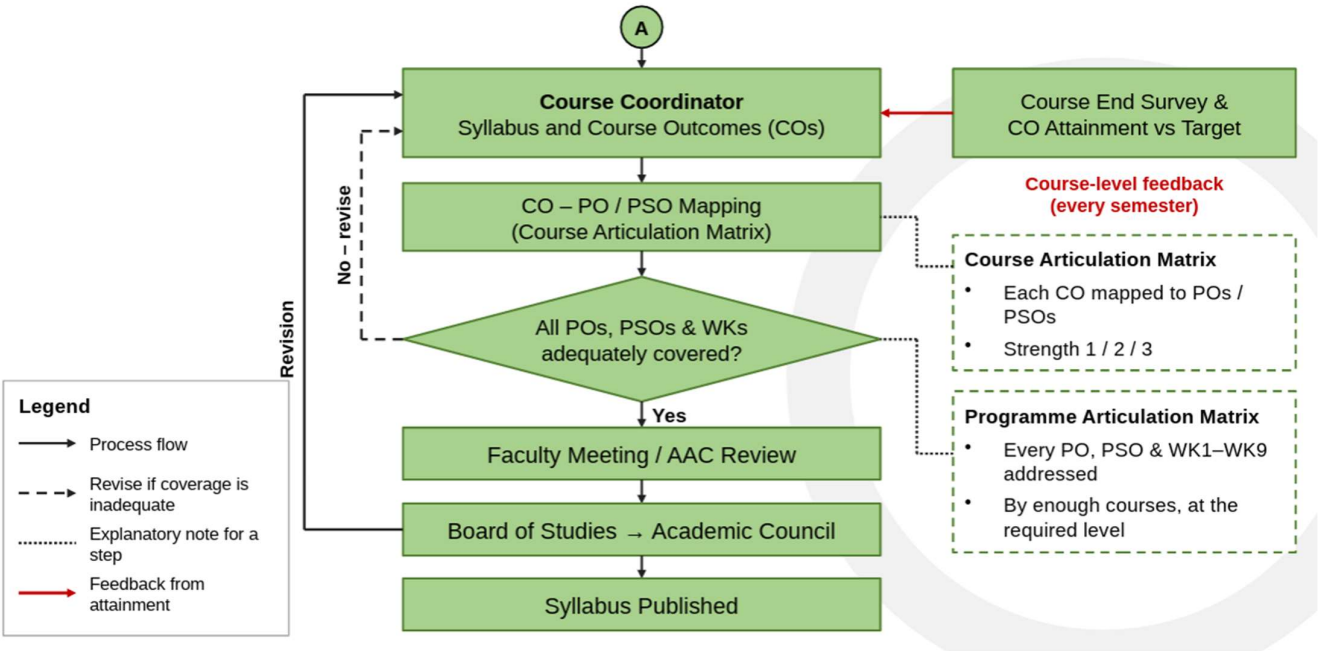
PO7 Ethics: Commit to professional ethics in managing software projects with financial aspects. Learn to use new technologies for cyber security and insulate customers from malware

PO8 Life-long learning: Change management skills and the ability to learn, keep up with contemporary technologies and ways of working.

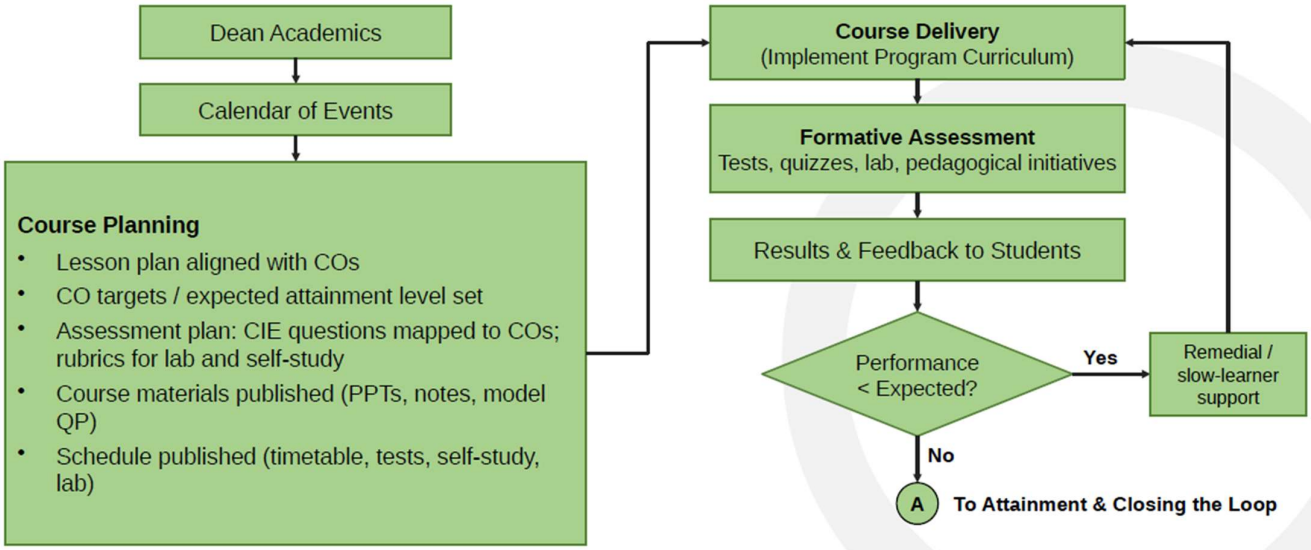
CURRICULUM DESIGN – SCHEME FORMULATION



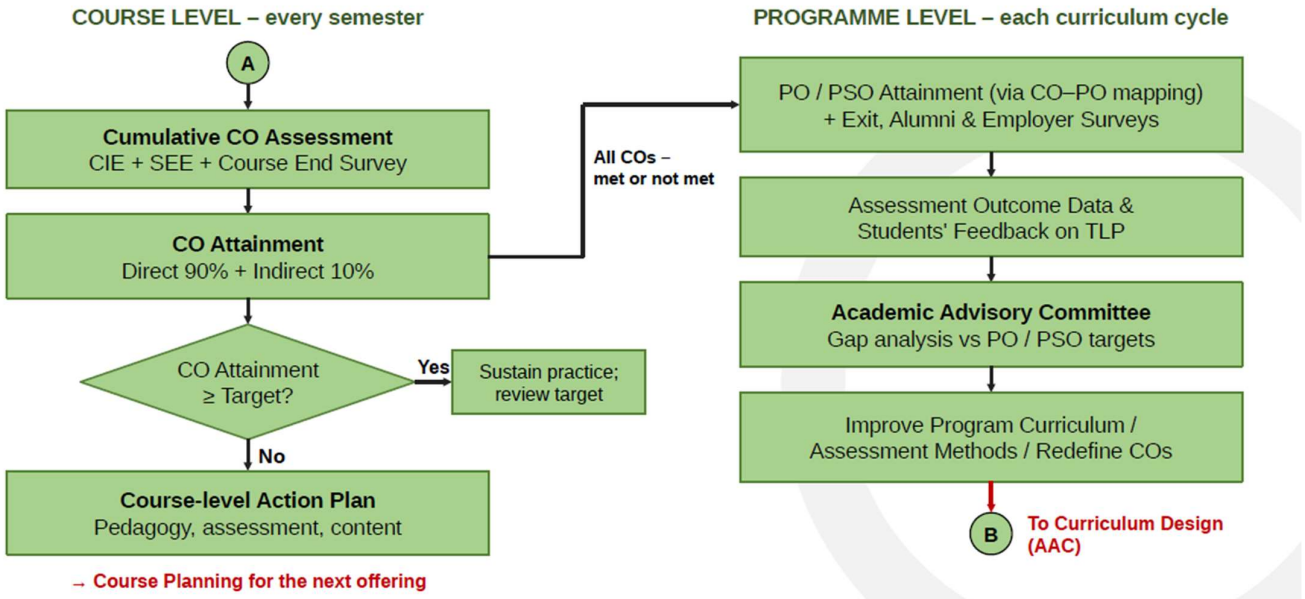
CURRICULUM DESIGN – SYLLABUS FORMULATION



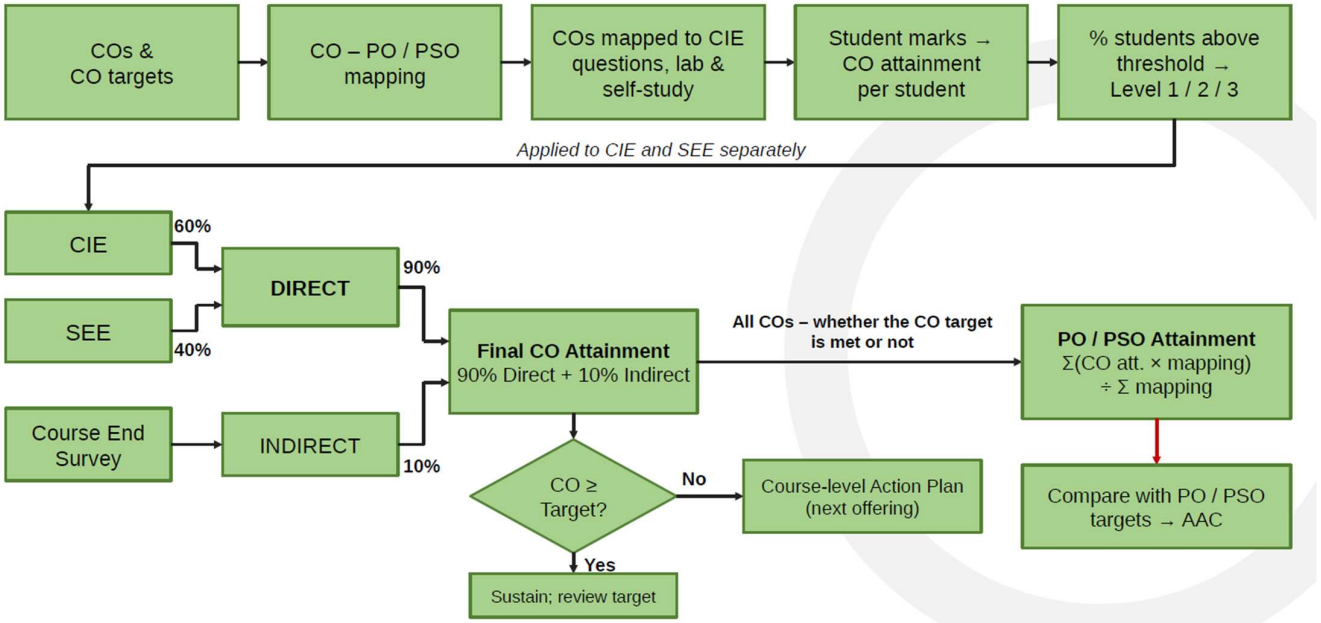
ACADEMIC PLANNING & DELIVERY



ATTAINMENT & CLOSING THE LOOP



CO & PO / PSO ATTAINMENT COMPUTATION



KNOWLEDGE & ATTITUDE

- **WK1:** A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
- **WK2:** Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
- **WK3:** A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
- **WK4:** Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
- **WK5:** Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
- **WK6:** Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.
- **WK7:** Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
- **WK8:** Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
- **WK9:** Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

INNOVATIVE TEAMS OF RVCE

Ashwa Mobility Foundation (AMF): Designs and fabricates Formula-themed race cars and mobility solutions to address urban transportation issues.

Astra Robotics Team: Focuses on designing and building application-specific robots.

Coding Club: Helps students gain coding skills and succeed in competitions like GSoC and ACM-ICPC.

Frequency Club Team: Works on software and hardware, emphasizing AI and Machine Learning.

Team Garuda: Develops a supermileage urban concept electric car and E-mobility products.

Team Jatayu: Builds low-cost UAVs with autonomous capabilities for various tasks.

Team Antariksh: Focuses on space technology and the development of operational rockets.

Team Chimera: Builds a Formula Electric Car through R&D in E-Mobility.

Helios Racing Team: Designs and tests All-Terrain Vehicles, participating in SAE's BAJA competitions.

Team Krushi: Creates low-cost farming equipment to assist farmers in cultivation and harvesting.

Team Vyoma: Designs and tests radio-controlled aircraft and UAVs.

Team Dhruva: Engages in astronomy-related activities and collaborates on projects with organizations like ICTS and IIA.

Ham Club: Promotes Amateur Radio and explores technical innovations in communications, especially for disaster response.

Chitrak : It is the official electric bike team of RV College of Engineering (RVCE) in Bangalore

Anoraniya:A Quantum based technical club initiated by students. The club is dedicated to addressing various technical challenges in quantum technologies and is on the verge of delivering some fascinating results.

Accelerate Club: organizes and participates in hackathons, bootcamps, workshops, and student-led innovation drives.

Cultural Activity Teams

- ALAAP
- Studio Zero
- CARV English
- CARV Hindi
- CARV Kannada
- Sattva
- F6.3 Photography club
- RAAG
- Rotaract Club of RVCE
- DebSoc
- QuizCorp RV
- Footprints
- Evoke
- TEDXRVCE

NSS of RVCE

NCC of RVCE



VISION

Leadership in Quality Technical Education, Interdisciplinary Research & Innovation, with a Focus on Sustainable and Inclusive Technology



MISSION

- To deliver outcome based Quality education, emphasizing on experientiallearning with the state of the art infrastructure.
- To create a conducive environment for interdisciplinary research and innovation.
- To develop professionals through holistic education focusing on individual growth, discipline, integrity, ethics and social sensitivity.
- To nurture industry-institution collaboration leading to competency enhancement and entrepreneurship.
- To focus on technologies that are sustainable and inclusive, benefiting all sections of the society.



QUALITY POLICY

Achieving Excellence in Technical Education, Research and Consulting through an Outcome Based Curriculum focusing on Continuous Improvement and Innovation by Benchmarking against the global Best Practices.



CORE VALUES

Professionalism, Commitment, Integrity, Team Work, Innovation